Observable Equivalence and Relational Invariants in Recursive Generative Systems

A Quotient Obstruction and Refinement-Compatibility Theorem
in the UNNS Substrate

UNNS Research Collective

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Abstract

Two generated structures can have the same observable representation and still interact differently with a third. We study this phenomenon in a deliberately minimal algebraic setting: a linear *route space* V, a linear observation map Π, and a bilinear interaction pairing B. We prove, with all hypotheses explicit, that $\Pi(X) = \Pi(X') \not\Rightarrow B(X,Y) = B(X',Y)$ in general, and we characterise exactly when the implication holds: *B factors through Π if and only if* $\ker\Pi \subseteq \mathrm{Rad}_L(B) \cap \mathrm{Rad}_R(B)$ (Observable Quotient Criterion). We introduce the *relational deficit* $\delta(\Pi, B) = \dim\ker\Pi/(\ker\Pi \cap \mathrm{Rad}\, B)$ and show that it is the least number of additional observables that restores interaction equivalence. We then separate *joint* refinement invariance, $B(TX,TY) = B(X,Y)$, from *independent* refinement invariance, $B(TX,Y) = B(X,Y) = B(X,TY)$: the former is compatible with arbitrarily large deficit (an explicit hidden-sector model, and the Legendre family $-P_\ell(\mathbf{a}\cdot\mathbf{b})$ with $\delta = 2\ell+1$ for $\ell \geq 2$), while under the hypothesis that refinement differences generate the observable-null directions the latter is equivalent to factorisation (Refinement-Compatibility Theorem). A final layer adds a transformation operator F and the *generative response profile* $G(X) = \big((\Pi F^n X)_n, (B(F^n X, \eta))_{n,\eta}\big)$: profile equivalence is governed by an unobservable subspace of linear systems theory, lies between route identity and observable identity, and identifies mechanisms only up to a gauge acting on the observable-null directions. The programme this suggests is stated as a clearly labelled *proposed* principle, with a record of which of its clauses are proved and which remain open. The tools are standard (universal property of quotients, radicals of bilinear forms, coinvariants, observability); the contribution is their unified treatment in a generative-route setting, the joint/independent distinction, the deficit, and worked, machine-checked counterexamples. For canonical UNNS the manuscript proves one consequence outright (the nest value $M \perp N$ does not determine the generated occurrence; the value 72 has three routes), states the rest as consequences conditional on an independently specified interaction, and records the construction of that interaction for a genuine UNNS mechanism as the open obligation. Apart from that consequence, everything is a statement about a *proposed mathematical realisation*; none is a theorem of canonical UNNS grammar, and no physical claim is made.

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1 Introduction

1.1 The question

A recursive generative system produces structures by rules: routes are composed, constrained by admissibility, and reduced to canonical forms. What an observer *sees* of a generated structure is a projection of it. The projection may be coarse; two structurally distinct routes can have the same observable representation. This is unremarkable as a fact about a single structure. The question studied here concerns *relations*: if two routes X, X' share an observable representation, must they stand in the same relations to every other route?

We formalise "relation" as a bilinear interaction pairing $B(X, Y)$ between routes, and "observable representation" as a linear map Π. Writing $\Pi(X) = \Pi(X')$ for observable equivalence and $B(X, Y) = B(X', Y)$ for all Y (together with the corresponding statement in the second slot) for interaction equivalence, the central point of this paper is

$$\boxed{\Pi(X) = \Pi(X') \quad \not\Rightarrow \quad B(X, Y) = B(X', Y) \ \ \forall Y} \tag{1}$$

as a general principle, together with a sharp characterisation of the situations in which the implication does hold. Equation (1) says that an observable representation need not be a complete description of a generative system's relational structure.

A second question, added in this version, is what a generative description can then legitimately claim. A recursive substrate need not reveal an invisible substructure by declaration; it must show how generative constraints become observable consequences, that is, how a structure *responds*, transforms, persists and interacts. Section 8 makes this precise in the same minimal setting.

1.2 The smallest instance

Let $V = \mathbb{R}^2$ with coordinates $X = (x, y)$, observation $\Pi(x, y) = x$ and pairing $B(X, Y) = xx' + yy'$. Take $X = (1, 1)$, $X' = (1, -1)$ and $Y = (0, 1)$. Then $\Pi X = \Pi X' = 1$, yet $B(X, Y) = 1$ while $B(X', Y) = -1$. The refinement $T(x, y) = (x, -y)$ preserves Π and satisfies $B(TX, TY) = B(X, Y)$ for all X, Y, but $B(TX, Y) \neq B(X, Y)$ in general. The remainder of the paper is, in essence, an exact account of what is happening in this example, why it persists in large and structured settings (hidden-sector models, rotationally covariant families, permutation modules), and which hypotheses rule it out.

1.3 Results

Three results form the core, and a fourth layer extends them from structure to response.

Result I (Observable Quotient Criterion, Thm. 4.1). A bilinear B factors through Π iff $\ker \Pi \subseteq \mathrm{Rad}_L(B) \cap \mathrm{Rad}_R(B)$. A quantitative companion (Prop. 4.7) gives the relational deficit δ.

Result II (Joint refinement invariance does not suffice, §6.1). In the route space $\mathbb{R}^3 \oplus \mathbb{R}^3$ the refinement $T(s, h) = (s, -h)$ preserves the observable component s and reverses the internal component h; the pairing $B(X, Y) = -\frac{1}{2}(s_X \cdot s_Y + h_X \cdot h_Y)$ is unchanged under *simultaneous* refinement of both arguments, yet $\delta = 3$.

Result	Content	Status
Thm. 4.1	B descends to Π iff $\ker \Pi \subseteq \mathrm{Rad}_L \cap \mathrm{Rad}_R$	standard (universal property of quotients); stated for arbitrary vector spaces
Prop. 4.7	deficit δ; bounds; minimal auxiliary observables	elementary; formulated here
Prop. 5.1	hidden-sector model: joint invariance, $\delta = 3$	explicit construction
Thm. 5.3	Legendre family: $\delta(B_\ell) = 2\ell + 1$, $\ell \geq 2$	short proof via the addition theorem
Prop. 6.2	involutive refinement $\Rightarrow$ block structure	elementary
Prop. 6.3	joint vs. independent invariance as diagonal vs. factorwise coinvariants	standard algebra, applied here
Thm. 7.1	refinement-compatibility (independent invariance $\Leftrightarrow$ factorisation)	elementary corollary of Thm. 4.1; **the technical point of the paper is the hypothesis structure**
Prop. 8.3, Cor. 8.4	profile congruence S_G; profile equivalence between route and observable identity	standard (unobservable subspace [6]); applied here
Props. 8.8–8.12	gauge non-identifiability; probe count ρ_L; restriction to admissible routes	elementary; formulated here
Princ. 8.14	generative characterisation	**proposal, not a theorem**

Table 1: Results and their status. None is claimed to be unprecedented as mathematics; the contribution is the unified generative-route treatment and the explicit joint/independent distinction.

Result III (Refinement-Compatibility Theorem, Thm. 7.1). If $\Pi T_i = \Pi$ and $\ker \Pi = \mathrm{span}\{X - T_i X\}$, then *independent* invariance $B(T_i X, Y) = B(X, Y) = B(X, T_i Y)$ holds iff B factors through Π.

Result IV (Generative response profiles, §8). With a transformation operator F, profile equivalence is $X - X' \in S_G$, the largest F-invariant subspace of $\ker \Pi \cap \mathrm{Rad}_L(B)$ (Prop. 8.3); it lies between route identity and observable identity (Cor. 8.4); mechanisms are identifiable only up to a gauge (Prop. 8.8); and the proposed Principle 8.14 is separated from what is proved.

In addition, §5.2 proves that the rotationally covariant, sign-reversing pairings $B_\ell(r(\mathbf{a}), r(\mathbf{b})) = -P_\ell(\mathbf{a} \cdot \mathbf{b})$ all have a finite, explicitly computed deficit $2\ell+1$ for $\ell \geq 2$ while B_1 factors through the observable vector $\mathbf{a}$.

1.4 Provenance and scope

The manuscript separates three kinds of statement, and marks them. *Adopted* items are vocabulary and constraints taken from existing UNNS project documents (§2.1); *introduced* items are the formal structures of this paper (linear route spaces, observation maps, pairings, refinement operators); *standard* items are classical linear algebra. All formal results are theorems about the introduced structures. We do *not* claim that canonical UNNS grammar entails any of the hypotheses used, and we do not claim a physical interpretation. Where the question of whether UNNS recursion satisfies a hypothesis arises, it is recorded as an open problem (§11).

1.5 Origin of the question

The problem arose from an attempt to derive a correlation law from UNNS-style joint generative structure (GCT programme). The preregistered v001 audit returned a no-go under Bell-local factorisation [13, 18]; v002 proved a binary joint-event representation and then showed, with an explicit family of kernels sharing all imposed symmetries, that those symmetries and operational no-signalling *underdetermine* the correlation kernel [14, 16]. A route-composition analysis then isolated the mathematical question treated here: whether route bilinearity together with refinement symmetry forces the interaction to depend only on observable projections. It does not, and that negative answer, together with the exact conditions under which it becomes positive, is the subject of this paper. We make no use of quantum probabilities, and nothing below is a statement about quantum mechanics, Bell's inequality [2, 4] or the Tsirelson bound [12]; these are mentioned once more only in the limitations (§11.2).

1.6 Organisation

Section 2 defines route systems and separates adopted from introduced structure. Section 3 constructs observable quotients and the three equivalences. Section 4 proves the quotient criterion and the deficit. Section 5 gives the obstruction examples. Sections 6 and 7 treat joint versus independent refinement invariance and prove the compatibility theorem. Section 8 adds the transformation operator, the response profile, identifiability and the proposed principle. Section 9 documents the deterministic verification, §10 discusses implications for UNNS, and §11 states limitations and open problems.

2 Generative Route Systems

2.1 What is adopted and what is introduced

The GCT v002 construction describes a candidate generative grammar as an alphabet Σ, a generation rule R, constraints C and an ordering O, and uses *admissibility* in the weak sense that each nonzero joint event is allowed [15]. That document is explicit that the grammar is a candidate and not canonical UNNS. We adopt its vocabulary (generation, constraints, admissibility, ordering) and its caution. Operator XVI (Closure) is described in Chamber XVI as an implementation of a Helmholtz decomposition that seals flux divergence [19]; we use it only as an illustration of a projection operator in §10.4.

2.2 Linear route systems

Definition 2.1 (Linear route system)**.** A *linear route system* is a quadruple $\mathfrak{S} = (V, \Pi, B, \mathcal{T})$ where

(R1) V is a real vector space (the *route space*), whose elements are finite signed combinations $X = \sum_i \alpha_i r_i$ of elementary routes r_i modulo the declared canonicalisation relations; *generative identity* is equality in V;

(R2) $\Pi\colon V \to O$ is a surjective linear map onto a real vector space O (the *observation map*);

(R3) $B\colon V \times V \to \mathbb{R}$ is bilinear (the *interaction pairing*);

(R4) $\mathcal{T}$ is a (possibly empty) set of linear maps $T\colon V \to V$ (the *refinements*).

Ingredient	Role here	Origin
Routes, generation, ordering	informal vocabulary for what V models	adopted [15]
Admissibility (constraints C)	declared constraints on the pairing (§2.3)	adopted vocabulary, *re-used in a stronger sense*
Closure as projection	example of an idempotent observation-like map (§10.4)	adopted [19, 17]
Mechanism $M = (\Sigma, R, C, O)$	ingredients, rules, constraints, ordering (§8)	adopted vocabulary [15]
Nest operation $M \perp N$, generated occurrence, five admissibility criteria	the canonical objects of §10.1	adopted [22]
Linear route space V	formal signed combinations of elementary routes	introduced
Observation map Π	linear map $V \to O$ onto an observable space	introduced
Interaction pairing B	bilinear map $V \times V \to \mathbb{R}$	introduced
Refinement operators T	linear maps with $\Pi T = \Pi$	introduced
Radicals, deficit, compatibility	analytic tools of this paper	introduced / standard
Transformation operator F, response profile G, protocol, S_G	the response layer (§8)	introduced; proposed formal extension, not canonical UNNS

Table 2: Separation of adopted and introduced structure.

We write $K = \ker \Pi$ for the space of *observable-null* routes. No finiteness, symmetry or topology is assumed unless stated.

Remark 2.2 (Modelling choices)**.** Linearity of V expresses that finite signed route collections compose; bilinearity of B expresses that interactions between disjoint route contributions add. Both are *hypotheses of the proposed model*. Route addition determines how interactions combine; it does not determine the elementary pairings $B(r_i, r_j)$. This under-determination is exactly what the examples of §5 exploit.

Example 2.3 (Free route spaces)**.** For a set E of elementary routes let $F(E)$ be the free real vector space on E. Any assignment $\Pi(r) \in O$ on generators extends uniquely to a linear Π, and any assignment of values $B(r, r')$ extends uniquely to a bilinear B. The principal example is $E = S^2$: $r(\mathbf{a})$ is an elementary route associated with an oriented observable direction $\mathbf{a}$, $O = \mathbb{R}^3$ and $\Pi(r(\mathbf{a})) = \mathbf{a}$.

Example 2.4 (Route space with an internal sector)**.** Let $V = O \oplus H$, $\Pi(o, h) = o$. Then $K = 0 \oplus H$ is the internal sector. The model of §5.1 is a change of coordinates of $V = \mathbb{R}^3 \oplus \mathbb{R}^3$ into this form.

2.3 Admissibility of a pairing

GCT v002 uses "admissibility" to mean that each nonzero joint event is allowed [15]. In this paper we need the stronger, constraint-based notion.

Definition 2.5 (Admissibility profile)**.** An *admissibility profile* $\mathcal{C}$ is a list of declared constraints on a pairing. A pairing is *$\mathcal{C}$-admissible* if it satisfies them.

The profile used in §5.2 mirrors the symmetry conditions frozen in the GCT v002 lock (rotational covariance, swap symmetry, aligned anticorrelation, axis inversion) [14, 20]. The structural lesson is that a pairing may satisfy every constraint of a profile and still fail to descend to the observable quotient.

2.4 Refinements

Definition 2.6 (Observable-preserving refinement)**.** A linear $T\colon V \to V$ is an *observable-preserving refinement* if $\Pi \circ T = \Pi$.

Informally, T changes how a structure is internally realised without changing what is observed of it. Equivalently, $(\mathrm{id} - T)V \subseteq K$. Four invariance notions are distinguished:

$$\text{joint:} \quad B(TX, TY) = B(X, Y), \tag{2}$$

$$\text{left:} \quad B(TX, Y) = B(X, Y), \tag{3}$$

$$\text{right:} \quad B(X, TY) = B(X, Y), \tag{4}$$

$$\text{independent (separate):} \quad (3) \text{ and } (4). \tag{5}$$

Independent invariance implies joint invariance, since $B(TX, TY) = B(X, TY) = B(X, Y)$. The converse is false (§6.1).

3 Observable Quotients

3.1 Fibres of the observation map

Proposition 3.1 (Observable quotient)**.** *Let $\Pi\colon V \to O$ be linear and onto, $K = \ker \Pi$. The fibres of Π are the cosets $X + K$, and Π induces a linear isomorphism $\bar{\Pi}\colon V/K \to O$. Two routes are observably equivalent, $X \sim_\Pi X'$ (i.e. $\Pi X = \Pi X'$), iff $X - X' \in K$.*

Proof. $\Pi X = \Pi X' \iff \Pi(X - X') = 0$. The isomorphism is the first isomorphism theorem [8]. □

Lemma 3.2 (Universal property)**.** *A linear map $F\colon V \to W$ factors as $F = \bar{F} \circ \Pi$ for a (necessarily unique) linear $\bar{F}\colon O \to W$ iff $K \subseteq \ker F$.*

Proof. If $F = \bar{F}\Pi$ then $F(K) = \bar{F}(0) = 0$. Conversely if $K \subseteq \ker F$ then $F(X)$ depends only on $X + K$, so $\bar{F}(\Pi X) := F(X)$ is well defined, and linear because Π is onto and linear. □

3.2 Radicals

Definition 3.3 (Radicals)**.** For bilinear B on V define $\mathrm{Rad}_L(B) = \{X : \ B(X, Y) = 0 \ \forall Y\}$, $\mathrm{Rad}_R(B) = \{Y : \ B(X, Y) = 0 \ \forall X\}$, and the *two-sided radical* $\mathrm{Rad}(B) = \mathrm{Rad}_L(B) \cap \mathrm{Rad}_R(B)$.

These are subspaces. If B is symmetric (or antisymmetric) then $\mathrm{Rad}_L = \mathrm{Rad}_R = \mathrm{Rad}$. B induces a canonical nondegenerate pairing on $V/\mathrm{Rad}_L \times V/\mathrm{Rad}_R$.

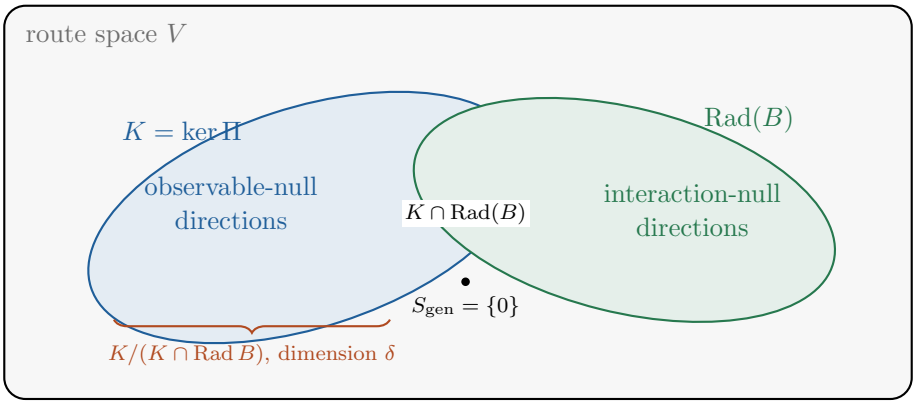


Figure 1: The three equivalences as three subspaces. Observable equivalence implies interaction equivalence exactly when the left ellipse lies inside the right one. The part of K that sticks out of $\mathrm{Rad}(B)$ is the relational content invisible to Π; its dimension is the relational deficit δ (Prop. 4.7).

3.3 Three equivalences

Definition 3.4 (Generative, observable and interaction equivalence)**.** For $X, X' \in V$:

- *generative equivalence* $X \sim_{\mathrm{gen}} X'$ iff $X = X'$ in V (same structural identity under the declared canonicalisation);
- *observable equivalence* $X \sim_{\Pi} X'$ iff $\Pi X = \Pi X'$;
- *interaction equivalence* $X \sim_B X'$ iff $B(X, Y) = B(X', Y)$ and $B(Y, X) = B(Y, X')$ for all $Y \in V$.

Each is a congruence of the form $X - X' \in S$ for a subspace S: $S_{\mathrm{gen}} = \{0\}$, $S_{\mathrm{obs}} = K$, $S_{\mathrm{int}} = \mathrm{Rad}(B)$.

Proposition 3.5 (Substitution principle)**.** *For two of the equivalences* $x, y \in \{\mathrm{gen}, \mathrm{obs}, \mathrm{int}\}$,

$$(\sim_x \Rightarrow \sim_y) \iff S_x \subseteq S_y.$$

Consequently: generative equivalence implies both others always; $\sim_{\Pi} \Rightarrow \sim_{\mathrm{gen}}$ *iff* $K = 0$*;* $\sim_B \Rightarrow \sim_{\mathrm{gen}}$ *iff* $\mathrm{Rad}(B) = 0$*;*

$$\sim_{\Pi} \Rightarrow \sim_B \iff K \subseteq \mathrm{Rad}(B), \qquad \sim_B \Rightarrow \sim_{\Pi} \iff \mathrm{Rad}(B) \subseteq K. \tag{6}$$

Proof. $\sim_x \Rightarrow \sim_y$ for all pairs means: $X - X' \in S_x \Rightarrow X - X' \in S_y$ for all X, X', i.e. $S_x \subseteq S_y$. □

Thus every implication between the three notions of identity is an inclusion between three subspaces of V, and the statement that none can be substituted for another without a compatibility theorem is the statement that three subspaces are in general position (Figure 1). The first criterion in (6) is Result I in disguise (Theorem 4.1).

4 Interaction Preservation

This section answers: for which observation maps Π does a bilinear pairing B *descend*, that is, satisfy $B(X, Y) = \bar{B}(\Pi X, \Pi Y)$ for a bilinear $\bar{B}$ on O?

4.1 The Observable Quotient Criterion

Theorem 4.1 (Observable Quotient Criterion — Result I)**.** *Let* $\Pi\colon V \to O$ *be linear and onto with kernel* K*, and let* $B\colon V \times V \to \mathbb{R}$ *be bilinear. The following are equivalent.*

(a) There is a bilinear $\bar{B}\colon O \times O \to \mathbb{R}$ *with* $B(X,Y) = \bar{B}(\Pi X, \Pi Y)$ *for all* X, Y*.*

(b) $\Pi X = \Pi X'$ *and* $\Pi Y = \Pi Y'$ *imply* $B(X,Y) = B(X',Y')$*.*

(c) $K \subseteq \mathrm{Rad}_L(B) \cap \mathrm{Rad}_R(B)$*.*

When they hold, $\bar{B}$ *is unique. No finiteness, symmetry or field-specific hypothesis beyond linearity is used.*

Proof. (a)⇒(b) is immediate. (b)⇒(c): let $k \in K$. Then $\Pi k = \Pi 0$, so for every Y, $B(k,Y) = B(0,Y) = 0$; hence $k \in \mathrm{Rad}_L$; symmetrically $k \in \mathrm{Rad}_R$. (c)⇒(a): for fixed Y the linear functional $B(\cdot, Y)$ vanishes on K, so by Lemma 3.2 there is a unique linear functional β_Y on O with $B(X,Y) = \beta_Y(\Pi X)$. For fixed $o \in O$ pick X with $\Pi X = o$; then $Y \mapsto \beta_Y(o) = B(X,Y)$ is linear and vanishes on K because $K \subseteq \mathrm{Rad}_R$; so it factors through Π as $\Pi Y \mapsto \bar{B}(o, \Pi Y)$. Linearity of $\bar{B}$ in o follows from that of B in X. Uniqueness: Π is onto. □

Status: standard (universal property of quotients applied to bilinear maps [8, 10, 9]); stated and proved here in the generality used below.

Corollary 4.2. *If* B *is symmetric or antisymmetric,* B *descends to* Π *iff* $K \subseteq \mathrm{Rad}(B)$*, i.e. iff the single condition* $B(k,\cdot) = 0$ *holds for all observable-null* k*.*

Proposition 4.3 (One-sided descent)**.** $B(X,Y) = \tilde{B}(\Pi X, Y)$ *for a bilinear* $\tilde{B}\colon O \times V \to \mathbb{R}$ *iff* $K \subseteq \mathrm{Rad}_L(B)$*; analogously for the second slot with* $\mathrm{Rad}_R(B)$*.*

Proof. Lemma 3.2 applied to $X \mapsto B(X,Y)$ for each Y. □

Proposition 4.3 shows that *both* radicals are needed in Theorem 4.1:

Example 4.4 (One-sided descent only)**.** $V = \mathbb{R}^2$, $\Pi(x,y) = x$, $B(X,Y) = x\,y'$. Then $K = \mathrm{span}\{(0,1)\}$, $B(k,\cdot) = 0$ for $k \in K$ so $K \subseteq \mathrm{Rad}_L$, but $B(\cdot,k) = x \neq 0$ so $K \not\subseteq \mathrm{Rad}_R$. B depends on its first argument only through Π, but on the second through y'; it does not descend.

4.2 Resolution ordering

Proposition 4.5 (Interaction quotient and resolution)**.** *Let* $q_L\colon V \to V/\mathrm{Rad}_L(B)$ *and* $q_R\colon V \to V/\mathrm{Rad}_R(B)$*. Then* B *factors canonically as* $B = \hat{B}(q_L\cdot, q_R\cdot)$ *with* $\hat{B}$ *nondegenerate, and*

$$B \text{ descends to } \Pi \iff q_L \text{ and } q_R \text{ both factor through } \Pi.$$

Proof. Nondegeneracy of $\hat{B}$ is the definition of the radicals. q_L factors through Π iff $K \subseteq \ker q_L = \mathrm{Rad}_L$ (Lemma 3.2); likewise for q_R. Apply Theorem 4.1. □

In words: *the observable resolution must dominate the interaction resolution.* If Π is coarser than q_L or q_R in the quotient-refinement order, some interaction information is invisible to Π.

4.3 Relational deficit

Definition 4.6 (Relational deficit)**.** $\delta(\Pi, B) := \dim\big(K/(K \cap \mathrm{Rad}(B))\big) \in \{0, 1, 2, \dots\} \cup \{\infty\}$ (cardinal dimension in general). For finite-dimensional data define also the left and right obstruction ranks $\rho_L = \dim K/(K \cap \mathrm{Rad}_L(B)) = \mathrm{rank}\big(X \mapsto B(X, \cdot)\big|_K\big)$ and ρ_R analogously.

Proposition 4.7 (Properties of the deficit)**.** *(i)* $\delta = 0$ *iff* B *descends to* Π.

(ii) $\max(\rho_L, \rho_R) \le \delta \le \rho_L + \rho_R$*; if* B *is symmetric or antisymmetric,* $\delta = \rho_L = \rho_R$.

(iii) (Matrix formula.) In bases, let the columns of K *span* $\ker \Pi$ *and let* $\mathsf{B}_{ij} = B(e_i, e_j)$*. Then*

$$\delta = \mathrm{rank}\begin{pmatrix} \mathsf{BK} \\ \mathsf{B}^\top\mathsf{K} \end{pmatrix}, \qquad \rho_L = \mathrm{rank}(\mathsf{B}^\top\mathsf{K}), \qquad \rho_R = \mathrm{rank}(\mathsf{BK}).$$

(iv) (Minimal completion.) Let $\Psi\colon V \to W$ *be linear.* B *descends to* $(\Pi, \Psi)\colon V \to O \oplus W$ *iff* $K \cap \ker \Psi \subseteq \mathrm{Rad}(B)$*. If* $\delta < \infty$*, the least* $\dim W$ *for which such a* Ψ *exists equals* δ.

Proof. (i) is Theorem 4.1 in the form $K \subseteq \mathrm{Rad}(B)$. (ii) Since $K \cap \mathrm{Rad} \subseteq K \cap \mathrm{Rad}_L$, $\rho_L \le \delta$; likewise $\rho_R \le \delta$. The map $K/(K \cap \mathrm{Rad}) \to K/(K \cap \mathrm{Rad}_L) \oplus K/(K \cap \mathrm{Rad}_R)$, $k \mapsto ([k], [k])$ is injective, giving $\delta \le \rho_L + \rho_R$. For symmetric/antisymmetric B all radicals coincide. (iii) Write $k = \mathsf{K}c$. Then $k \in \mathrm{Rad}_L$ iff $k^\top \mathsf{B} = 0$ iff $\mathsf{B}^\top\mathsf{K}c = 0$, and $k \in \mathrm{Rad}_R$ iff $\mathsf{BK}c = 0$. So $K \cap \mathrm{Rad}(B)$ corresponds to the null space of the stacked matrix and $\delta = \dim K - \dim \ker = \mathrm{rank}$. (iv) The kernel of (Π, Ψ) is $K \cap \ker \Psi$; apply Theorem 4.1. If $K \cap \ker \Psi \subseteq \mathrm{Rad}(B)$ then $\Psi|_K$ has kernel inside $K \cap \mathrm{Rad}$, so $\dim W \ge \dim K/(K \cap \mathrm{Rad}) = \delta$. Conversely let $\Psi_0\colon K \to K/(K \cap \mathrm{Rad})$ be the quotient map and extend it linearly to V (choose a complement of K); then $K \cap \ker \Psi = K \cap \mathrm{Rad}$ and $\dim W = \delta$. □

Status: elementary; the deficit and the minimal-completion statement are formulations introduced here. Verified numerically on 300 random instances (Appendix B).

Informally, δ counts the independent pieces of generative structure that are invisible to the observable representation yet visible to the interaction. In the example of §1.2, $K = \mathrm{span}\{(0, 1)\}$, $\mathrm{Rad}(B) = 0$, hence $\delta = 1$: one extra observable (y) is exactly what is needed.

4.4 Self-interaction and representatives

Two further statements explain why the obstruction cannot be detected from the wrong data, and why the pairing induced on observables is not canonical.

Proposition 4.8 (Self-interaction)**.** *Let* B *be symmetric and* $Q(X) = B(X, X)$*. Then* $Q(X) = \bar{Q}(\Pi X)$ *for some function* $\bar{Q}$ *on* O *iff* B *descends to* Π.

Proof. If B descends, $Q(X) = \bar{B}(\Pi X, \Pi X)$. Conversely let $k \in K$, $X \in V$, $t \in \mathbb{R}$. Then $0 = Q(X + tk) - Q(X) = 2tB(X, k) + t^2 B(k, k)$ for all t, so $B(X, k) = 0$ for all X and $B(k, k) = 0$. Hence $K \subseteq \mathrm{Rad}_R(B) = \mathrm{Rad}(B)$ by symmetry. □

The self-interaction is a function of the observable *only* if it is constant on every fibre $X + K$. A refinement T with $B(TX, TX) = B(X, X)$ (a consequence of joint invariance) tests constancy on the two-point set $\{X, TX\} \subseteq X + K$ only. In the hidden-sector model of §5.1, $Q(X + k) - Q(X) = -\frac{1}{2} h_k \cdot h_k - h_k \cdot h_X$, which vanishes at $k = -2h_X$ (the refinement partner, where $h_{X+k} = -h_X$) but not for general $k \in K$. Joint invariance is therefore the wrong test for observable adequacy.

Proposition 4.9 (Representatives)**.** *Let $\sigma\colon O \to V$ be a linear section of Π ($\Pi\sigma = \mathrm{id}_O$) and put $B_\sigma(o, o') = B(\sigma o, \sigma o')$.*

(a) Every other linear section is $\sigma' = \sigma + \kappa$ with $\kappa\colon O \to K$ linear, and

$$B_{\sigma'}(o, o') - B_\sigma(o, o') = B(\sigma o, \kappa o') + B(\kappa o, \sigma o') + B(\kappa o, \kappa o').$$

(b) If B descends, $B_\sigma = \bar{B}$ for every σ.

(c) If B is symmetric, $O \neq 0$, and B_σ does not depend on σ, then B descends.

In the hidden-sector model, the linear sections are $\sigma_H(s) = (s, Hs)$ in (s, h)-coordinates with $H \in \mathbb{R}^{3\times 3}$ arbitrary, and

$$B_{\sigma_H}(s, s') = -\tfrac{1}{2}\, s^\top (\mathsf{I} + H^\top H)\, s'.$$

Proof. (a) $\kappa = \sigma' - \sigma$ lands in K because $\Pi\sigma' = \Pi\sigma$; the formula is bilinearity. (b) is Theorem 4.1. (c) The hypothesis $O \neq 0$ is needed: for $V = \mathbb{R}$, $O = 0$, $B(x, y) = xy$ the only section is 0, so B_σ is trivially independent of σ, yet $K = V$ and $\mathrm{Rad}(B) = 0$. Now let $k \in K$, $0 \neq \varphi \in O^*$ (which exists since $O \neq 0$), and apply (a) with $\kappa = t\,\varphi(\cdot)k$. The coefficient of t^2 is $\varphi(o)\varphi(o')B(k, k)$, forcing $B(k, k) = 0$. The coefficient of t is $\varphi(o')B(\sigma o, k) + \varphi(o)B(\sigma o', k)$ (using symmetry); taking $o = o'$ with $\varphi(o) \neq 0$ gives $B(\sigma o, k) = 0$ on a set that spans O, hence for all o. Finally every $X \in V$ is $X = \sigma\Pi X + k'$ with $k' \in K$, and polarising $B(k, k) = 0$ on the subspace K gives $B(K, K) = 0$; so $B(X, k) = 0$ for all X and $K \subseteq \mathrm{Rad}(B)$. The displayed formula is the model-H pairing evaluated on $\sigma_H(s) = (s, Hs)$. □

Status: elementary. The model-H formula is reproduced numerically (Appendix B). Thus the pairing one would *read off* on observables by choosing a representative of each observable value is canonical exactly when the interaction descends. In the hidden-sector model every choice of H gives a legitimate but different observable pairing, from $-\frac{1}{2}s\cdot s'$ at $H = 0$ to arbitrarily large magnitudes; the observation map alone does not select among them.

5 The Quotient Obstruction

We now give three families in which the obstruction is present although every "natural" symmetry is respected.

5.1 The hidden-sector model

Let $V = \mathbb{R}^3 \oplus \mathbb{R}^3$ with elements $X = (u, v)$. Define the observable and internal components

$$s = u + v, \qquad h = u - v,$$

the observation map $\Pi(X) = s_X \in \mathbb{R}^3$, the pairing

$$B(X, Y) = -\tfrac{1}{2}\big(s_X \cdot s_Y + h_X \cdot h_Y\big), \tag{7}$$

and the refinement $T(u, v) = (v, u)$, which in the new coordinates reads

$$T(s, h) = (s, -h). \tag{8}$$

Proposition 5.1 (Properties of the hidden-sector model)**.** *(a) $B(X, Y) = -(u_X \cdot u_Y + v_X \cdot v_Y)$; B is symmetric, negative definite,* $\mathrm{Rad}(B) = 0$.

(b) $\Pi T = \Pi$, $h_{TX} = -h_X$ *(observable component unchanged, internal component reversed).*

(c) Joint invariance: $B(TX, TY) = B(X, Y)$.

(d) $B(TX, Y) = B(X, Y) + h_X \cdot h_Y$*; hence independent invariance fails.*

(e) $K = \{(u, -u)\} = \{s = 0\}$, $\dim K = 3$, *and* $\operatorname{span}\{X - TX\} = K$.

(f) $\delta(\Pi, B) = 3$. *In particular* B *does* not *descend to* Π*: for* $k = (u, -u)$ *and every* Y, $B(k, Y) = -\frac{1}{2} h_k \cdot h_Y = -u \cdot h_Y$.

Proof. (a) $s \cdot s' + h \cdot h' = (u+v)\cdot(u'+v') + (u-v)\cdot(u'-v') = 2(u \cdot u' + v \cdot v')$. (b) $s_{TX} = v + u = s_X$, $h_{TX} = v - u = -h_X$. (c) $B(TX, TY) = -\frac{1}{2}(s_X \cdot s_Y + (-h_X) \cdot (-h_Y))$. (d) $B(TX, Y) = -\frac{1}{2}(s_X \cdot s_Y - h_X \cdot h_Y) = B(X, Y) + h_X \cdot h_Y$. (e) $s = 0 \iff v = -u$. $X - TX = (u - v, v - u) \in K$ and every $(w, -w)$ arises with $u = w/2$, $v = -w/2$. (f) $K \cap \operatorname{Rad}(B) = 0$ as $\operatorname{Rad}(B) = 0$, so $\delta = \dim K = 3$; the displayed value of $B(k, Y)$ follows from (7) with $s_k = 0$, $h_k = 2u$. □

Example 5.2 (Concrete witness)**.** Let $\mathbf{e}_x = (1, 0, 0)$ and take $X = (\mathbf{e}_x, \mathbf{0})$, $X' = TX = (\mathbf{0}, \mathbf{e}_x)$. Then $\Pi X = \Pi X' = \mathbf{e}_x$, but with $Y = X$: $B(X, X) = -1$ and $B(X', X) = 0$. Moreover $B(X, X) = B(X', X') = -1$, so the two routes agree on observable value and on self-interaction, and differ in their mutual interaction with X.

This is exactly the structure of the minimal example of §1.2, with a three-dimensional hidden sector.

5.2 The Legendre family on the route space of directions

Let $V = F(S^2)$ with $\Pi(r(\mathbf{a})) = \mathbf{a} \in \mathbb{R}^3$ (Example 2.3), and let P_ℓ be the Legendre polynomial of degree ℓ. For $\ell \geq 1$ define B_ℓ on generators by

$$B_\ell\big(r(\mathbf{a}), r(\mathbf{b})\big) = -P_\ell(\mathbf{a} \cdot \mathbf{b}), \tag{9}$$

extended bilinearly. In particular $B_1(r(\mathbf{a}), r(\mathbf{b})) = -\mathbf{a} \cdot \mathbf{b}$ and $B_3 = -P_3(\mathbf{a} \cdot \mathbf{b})$ with $P_3(x) = \frac{1}{2}(5x^3 - 3x)$.

Theorem 5.3 (The Legendre family)**.** *(i) (Admissibility profile.)* B_ℓ *is symmetric, satisfies* $B_\ell(r(\mathbf{a}), r(\mathbf{a})) = -1$, *and* $-B_\ell$ *is positive semidefinite on* V*; it is covariant under rotations,*

$$B_\ell\big(r(R\mathbf{a}), r(R\mathbf{b})\big) = B_\ell\big(r(\mathbf{a}), r(\mathbf{b})\big), \qquad R \in O(3),$$

and, for odd ℓ, $B_\ell(r(-\mathbf{a}), r(\mathbf{b})) = -B_\ell(r(\mathbf{a}), r(\mathbf{b}))$.

(ii) B_1 *descends to* Π*:* $B_1(X, Y) = -\Pi X \cdot \Pi Y$*;* $\delta(\Pi, B_1) = 0$.

(iii) For every $\ell \geq 2$, B_ℓ *does not descend to* Π, *and* $\delta(\Pi, B_\ell) = \rho_L = \rho_R = 2\ell + 1$.

Proof. (i) Rotation invariance of $\mathbf{a} \cdot \mathbf{b}$; $P_\ell(1) = 1$; $P_\ell(-x) = (-1)^\ell P_\ell(x)$. Let $S_{\ell m}$, $m = 1, \ldots, 2\ell + 1$, be real orthonormal spherical harmonics of degree ℓ. The addition theorem [1] gives

$$P_\ell(\mathbf{a} \cdot \mathbf{b}) = \frac{4\pi}{2\ell + 1} \sum_{m=1}^{2\ell+1} S_{\ell m}(\mathbf{a}) S_{\ell m}(\mathbf{b}). \tag{10}$$

Define linear functionals $f_m \colon V \to \mathbb{R}$ by $f_m(r(\mathbf{a})) = S_{\ell m}(\mathbf{a})$. Then

$$B_\ell(X, Y) = -c_\ell \sum_m f_m(X) f_m(Y), \qquad c_\ell = \tfrac{4\pi}{2\ell+1} > 0, \tag{11}$$

so $-B_\ell(X,X) = c_\ell \sum f_m(X)^2 \geq 0$. (ii) $P_1(x) = x$ and $B_1(r(\mathbf{a}), r(\mathbf{b})) = -\Pi r(\mathbf{a}) \cdot \Pi r(\mathbf{b})$; both sides are bilinear and agree on generators. (iii) The f_m are linearly independent functionals on V because the $S_{\ell m}$ are linearly independent functions on S^2 (evaluate on generators). Consequently $F = (f_1, \dots, f_{2\ell+1})\colon V \to \mathbb{R}^{2\ell+1}$ is onto, and by (11), $B_\ell(X, \cdot) = 0 \iff \sum_m f_m(X) f_m = 0 \iff F(X) = 0$; so $\operatorname{Rad}(B_\ell) = \ker F$ and

$$\delta = \dim K/(K \cap \ker F) = \dim F(K).$$

Suppose $F(K) \neq \mathbb{R}^{2\ell+1}$. Then there is $c \neq 0$ with $c \cdot F(X) = 0$ for all $X \in K$, i.e. the functional $\varphi = \sum c_m f_m$ vanishes on $K = \ker \Pi$. By Lemma 3.2, $\varphi = w \circ \Pi$ for some $w \in \mathbb{R}^3$. Evaluating on generators, $\sum_m c_m S_{\ell m}(\mathbf{a}) = w \cdot \mathbf{a}$ for all $\mathbf{a} \in S^2$. The left side is a spherical harmonic of degree ℓ, the right a harmonic of degree 1; for $\ell \neq 1$ these are orthogonal in $L^2(S^2)$, so both vanish, and $c = 0$ by independence: contradiction. Hence $F(K) = \mathbb{R}^{2\ell+1}$ and $\delta = 2\ell + 1$. Symmetry of B_ℓ gives $\rho_L = \rho_R = \delta$ (Prop. 4.7(ii)). □

Example 5.4 (Explicit witness, $\ell = 3$)**.** Let $\mathbf{n} = \frac{1}{2}\mathbf{e}_x + \frac{\sqrt{3}}{2}\mathbf{e}_y$ and

$$X = r(\mathbf{n}) - \tfrac{1}{2}\, r(\mathbf{e}_x) - \tfrac{\sqrt{3}}{2}\, r(\mathbf{e}_y).$$

Then $\Pi X = \mathbf{n} - \frac{1}{2}\mathbf{e}_x - \frac{\sqrt{3}}{2}\mathbf{e}_y = 0$, so $X \in K$. Using $P_3(\frac{1}{2}) = -\frac{7}{16}$, $P_3(1) = 1$, $P_3(0) = 0$:

$$B_1\big(X, r(\mathbf{e}_x)\big) = -\tfrac{1}{2} + \tfrac{1}{2} = 0, \qquad B_3\big(X, r(\mathbf{e}_x)\big) = \tfrac{7}{16} + \tfrac{1}{2} = \tfrac{15}{16}.$$

Hence $X \notin \operatorname{Rad}_L(B_3)$ although $\Pi X = 0$. The same X gives, for $\ell = 5$ ($P_5(\frac{1}{2}) = \frac{23}{256}$), $B_5(X, r(\mathbf{e}_x)) = \frac{105}{256}$.

Example 5.5 (A four-route table)**.** Restrict to the four elementary routes $r_1 = r(\mathbf{e}_x)$, $r_2 = r(\mathbf{e}_y)$, $r_3 = r(\mathbf{e}_z)$, $r_4 = r(\mathbf{n})$, with $\mathbf{n} = (\frac{1}{2}, \frac{\sqrt{3}}{2}, 0)$. Then the route space is $V_4 = \mathbb{R}^4$, Π has matrix $(\mathbf{e}_x\ \mathbf{e}_y\ \mathbf{e}_z\ \mathbf{n})$, $K = \operatorname{span}\{k\}$ with $k = (-\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0, 1)^\top$, and the Gram matrices of B_1 and B_3 are

$$G_1 = -\begin{pmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{\sqrt{3}}{2} \\ 0 & 0 & 1 & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 & 1 \end{pmatrix}, \qquad G_3 = \begin{pmatrix} -1 & 0 & 0 & \frac{7}{16} \\ 0 & -1 & 0 & -\frac{3\sqrt{3}}{16} \\ 0 & 0 & -1 & 0 \\ \frac{7}{16} & -\frac{3\sqrt{3}}{16} & 0 & -1 \end{pmatrix},$$

using $P_3(\frac{\sqrt{3}}{2}) = \frac{3\sqrt{3}}{16}$. Then

$$G_1 k = 0, \qquad G_3 k = \Big(\tfrac{15}{16},\ \tfrac{5\sqrt{3}}{16},\ 0,\ -\tfrac{15}{16}\Big)^\top \neq 0,$$

so $\delta = 0$ for B_1 and $\delta = 1$ for B_3 on this four-route space: the single observable-null route k interacts nontrivially with r_1, r_2 and r_4 under B_3 and with nothing under B_1. Every entry is verified in exact arithmetic over $\mathbb{Q}(\sqrt{3})$ (Appendix B).

Remark 5.6 (Antipodal quotient)**.** If one prefers a route space in which $r(-\mathbf{a}) = -r(\mathbf{a})$, replace $F(S^2)$ by its quotient by $\operatorname{span}\{r(\mathbf{a}) + r(-\mathbf{a})\}$. For odd ℓ, Π and B_ℓ are well defined on the quotient, the witness of Example 5.4 lives there unchanged, and the proof of Theorem 5.3(iii) goes through verbatim because $S_{\ell m}$ are odd for odd ℓ. The obstruction is therefore not an artefact of freeness.

	$\ell = 1$	$\ell = 3$	$\ell = 5$
$C = B_\ell(r(\mathbf{a}), r(\mathbf{b}))$ at $60°$ $(\mathbf{a}\cdot\mathbf{b} = \frac{1}{2})$	$-\frac{1}{2}$	$+\frac{7}{16}$	$-\frac{23}{256}$
$\frac{1}{4}(1 + C)$	$\frac{1}{8} = 0.125$	$\frac{23}{64} = 0.359375$	$\frac{233}{1024}$
$B_\ell(X, r(\mathbf{e}_x))$ for the $X \in K$ of Ex. 5.4	0	$\frac{15}{16}$	$\frac{105}{256}$
$\delta(\Pi, B_\ell)$	0	7	11

Table 3: Values (exact fractions, reproduced by the verification script of §9). The second row applies the binary joint-event representation $P(A, B \mid \mathbf{a}, \mathbf{b}) = \frac{1}{4}(1 + AB\,C)$ proved in GCT v002 [14] purely as arithmetic on the first row; no probabilistic claim is made in this paper.

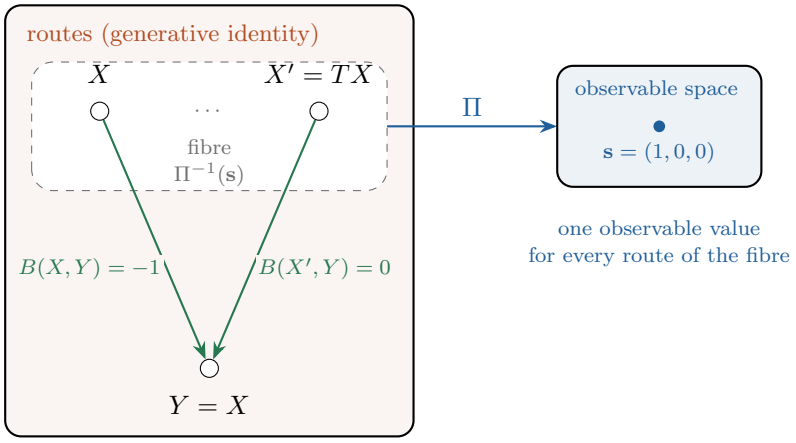


Figure 2: Many routes, one observable. The hidden-sector model of §5.1 (Example 5.2): X and $X' = TX$ lie in one fibre of Π and have the same self-interaction $B(X, X) = B(X', X') = -1$, but different pairings with $Y = X$.

Remark 5.7 (Relation to the GCT v002 underdetermination theorem)**.** The v002 lock freezes the conditions $C(\mathbf{a}, \mathbf{b}) = c(a - b)$, $c(-d) = c(d)$, $c(0) = -1$, $c(d + \pi) = -c(d)$ together with $|c| \leq 1$ [20, 14]. For planar directions the kernels $c_\ell(d) = -P_\ell(\cos d)$, ℓ odd, satisfy all of them (checked in Appendix B for $\ell = 3$). In route-space language, *what distinguishes B_1 from B_3 among these admissible pairings is precisely whether the pairing descends to the observable vector*: $\delta = 0$ versus $\delta = 7$. The descent condition $K \subseteq \mathrm{Rad}(B)$ is therefore a *selection criterion* within the admissible class. It is not a derivation: whether UNNS recursion supplies it independently is open (§11).

5.3 Illustration: many routes, one observable

6 Refinement and Relational Invariance

The hidden-sector model of §5.1 has a refinement T for which the interaction is invariant under the simultaneous refinement of both arguments. This section isolates what that does and does not imply.

6.1 Joint refinement invariance does not eliminate hidden interactions

Theorem 6.1 (Result II)**.** *There exist a linear route system* $(V, \Pi, B, \{T\})$ *with* $\Pi T = \Pi$, $\operatorname{span}\{X - TX\} = \ker \Pi$*, and* B *symmetric, nondegenerate and jointly invariant,* $B(TX, TY) = B(X, Y)$*, such that* B *does not descend to* Π *and* $\delta(\Pi, B) = \dim \ker \Pi > 0$*.*

Proof. The hidden-sector model, Proposition 5.1(b)–(f). □

Thus joint refinement invariance need not eliminate interactions involving individually unobserved structural distinctions: $B(TX, Y) - B(X, Y) = h_X \cdot h_Y$ is nonzero whenever the internal components of X and Y overlap, even though the same distinction cancels when *both* routes are refined.

6.2 Involutive refinements: a block decomposition

Proposition 6.2 (Involutive refinement)**.** *Let* T *be a refinement (*$\Pi T = \Pi$*) with* $T^2 = \mathrm{id}$*. Put* $V_\pm = \{X : TX = \pm X\}$*, so* $V = V_+ \oplus V_-$*, and assume* $\ker \Pi = V_-$ *(so that* $\Pi|_{V_+}$ *is injective). Then:*

(a) B *is jointly invariant iff* $B(V_+, V_-) = B(V_-, V_+) = 0$*, i.e.* $B = B|_{V_+} \oplus B|_{V_-}$*.*

(b) B *is left-invariant iff* $V_- \subseteq \mathrm{Rad}_L(B)$*; right-invariant iff* $V_- \subseteq \mathrm{Rad}_R(B)$*.*

(c) Under joint invariance, B *descends to* Π *iff* $B|_{V_- \times V_-} = 0$*. In that case* $B = B|_{V_+}$ *and* $\delta = 0$*; in general* $\delta = \dim V_- / \big(V_- \cap \mathrm{Rad}(B|_{V_-})\big)$*.*

Proof. We use that the characteristic of $\mathbb{R}$ is not 2. (a) For $x \in V_+$, $y \in V_-$: joint invariance gives $B(x, y) = B(Tx, Ty) = -B(x, y)$, hence $B(x, y) = 0$; similarly for $B(y, x)$. Conversely if B is block diagonal, write $X = x_+ + x_-$, $Y = y_+ + y_-$ and compute $B(TX, TY) = B(x_+, y_+) + B(x_-, y_-) = B(X, Y)$. (b) $B(TX, Y) - B(X, Y) = -2B(x_-, Y)$. (c) $K = V_- \subseteq \mathrm{Rad}(B)$ iff $B(V_-, V_-) = 0$ in view of the block structure; the formula for δ follows because $B(V_-, V_+) = 0$. □

The hidden-sector model is the case $V_+ = \{h = 0\}$, $V_- = \{s = 0\}$ with $B|_{V_-} = -\frac{1}{2} h \cdot h'$ nondegenerate. Joint invariance is exactly the statement that no interaction couples the observable and the internal sector; it says nothing about the internal sector's *own* interaction, which survives all refinements of both arguments.

6.3 Diagonal versus factorwise coinvariants

Let a group G act on V by refinements (so $\Pi g = \Pi$ for all $g \in G$) and write $V_G = V / \operatorname{span}\{X - gX\}$ for the coinvariants [3].

Proposition 6.3 (Joint versus independent invariance)**.** *(a) Bilinear forms that are* independently *invariant (left and right) under* G *are exactly the bilinear forms on* V *that factor through* $V_G \times V_G$*, i.e.* $(V_G \otimes V_G)^*$*.*

(b) Bilinear forms that are jointly *invariant under* G *are exactly* $((V \otimes V)_G)^*$*, coinvariants for the diagonal action* $g \mapsto g \otimes g$*.*

(c) The natural surjection $(V \otimes V)_G \twoheadrightarrow V_G \otimes V_G$ *has kernel* N*, and the gap between joint and independent invariance is the space of functionals on* $(V \otimes V)_G$ *that do not vanish on* N*.*

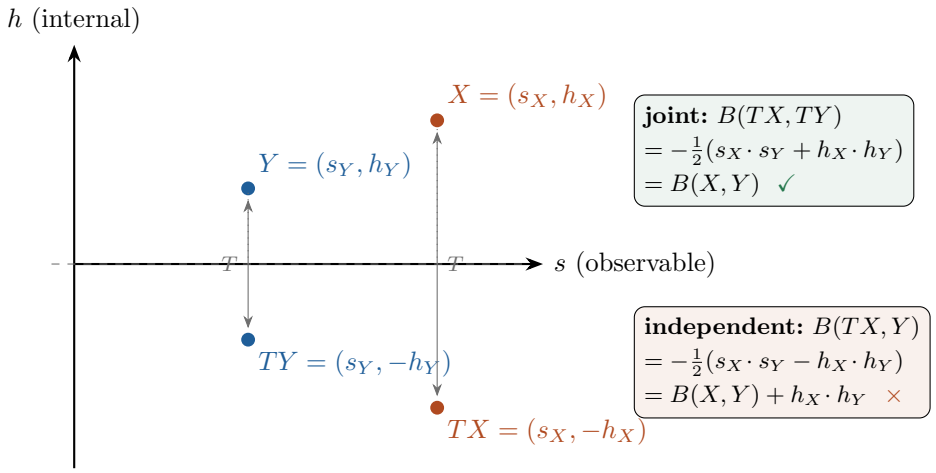


Figure 3: Refinement and relational preservation (one scalar internal coordinate drawn). T reflects the internal coordinate and fixes the observable one. Reflecting *both* routes preserves the interaction; reflecting only one changes it by $h_X \cdot h_Y$.

For $G = \mathbb{Z}_2 = \{1, T\}$ acting on $V = V_+ \oplus V_-$ (over $\mathbb{R}$): $V_G = V_+$, $(V \otimes V)_G = V_+ \otimes V_+ \oplus V_- \otimes V_-$, so $N = V_- \otimes V_-$.

Proof. (a) Left- and right-invariance say B vanishes on $\mathrm{span}\{X - gX\} \times V$ and $V \times \mathrm{span}\{X - gX\}$, i.e. B factors through $V_G \times V_G$ (universal property for each slot, as in Theorem 4.1). (b) Joint invariance says that the linear functional $\tilde{B}$ on $V \otimes V$ vanishes on $X \otimes Y - gX \otimes gY$. (c) The natural map $V \otimes V \to V_G \otimes V_G$ sends the span of $X \otimes Y - gX \otimes gY$ into the span of $\{X - gX\} \otimes V + V \otimes \{Y - gY\}$ because $X \otimes Y - gX \otimes gY = (X - gX) \otimes Y + gX \otimes (Y - gY)$; so it descends. For $\mathbb{Z}_2$: $T \otimes T$ acts by $+1$ on $V_\pm \otimes V_\pm$ and -1 on $V_+ \otimes V_-$ and $V_- \otimes V_+$, whose coinvariants vanish in characteristic $\neq 2$. □

For the hidden-sector model, $\dim N = \dim V_- \otimes V_- = 9$, and the hidden interaction $-\frac{1}{2} h \cdot h'$ is a functional on $(V \otimes V)_G$ that is supported on N, hence invisible to $V_G \otimes V_G$. This is the algebraic reason that the joint/independent distinction is real: *the diagonal coinvariants are strictly larger than the tensor square of the single-argument coinvariants.*

7 The Observable-Null Compatibility Theorem

7.1 Hypotheses

Let $\{T_i\}_{i \in I}$ be a family of refinements of (V, Π). We use two hypotheses.

(H1) Observable preservation. $\Pi T_i = \Pi$ for all i.

(H2) Kernel generation. $\ker \Pi = \mathrm{span}\{X - T_i X : \ X \in V, \ i \in I\}$.

(H1) implies $\mathrm{span}\{X - T_i X\} \subseteq \ker \Pi$; (H2) asserts that refinement differences exhaust the observable-null directions. Equivalently, Π is (isomorphic to) the quotient of V by the span of refinement differences: observable identity *is* identity up to refinement.

7.2 The theorem

Theorem 7.1 (Refinement-Compatibility Theorem — Result III)**.** *Assume (H1) and (H2). Let B be bilinear on V satisfying independent invariance for every $i \in I$:*

$$B(T_iX, Y) = B(X, Y), \qquad B(X, T_iY) = B(X, Y) \qquad (X, Y \in V).$$

Then $\ker \Pi \subseteq \mathrm{Rad}_L(B) \cap \mathrm{Rad}_R(B)$*; consequently B factors through Π: $B(X, Y) = \bar{B}(\Pi X, \Pi Y)$ for a unique bilinear $\bar{B}$. Conversely, (H1) alone shows that any B which factors through Π is independently invariant under every T_i. Hence, under (H1)–(H2),*

$$B \text{ independently invariant} \iff B \text{ factors through } \Pi \iff \ker \Pi \subseteq \mathrm{Rad}(B).$$

Proof. For each X and i, left-invariance gives $B(X - T_iX, Y) = 0$ for all Y, i.e. $X - T_iX \in \mathrm{Rad}_L(B)$; right-invariance gives $B(Y, X - T_iX) = 0$, i.e. $X - T_iX \in \mathrm{Rad}_R(B)$. Radicals are subspaces, so by (H2) $\ker \Pi = \mathrm{span}\{X - T_iX\} \subseteq \mathrm{Rad}_L \cap \mathrm{Rad}_R$. Apply Theorem 4.1. For the converse, if $B = \bar{B}(\Pi\cdot, \Pi\cdot)$ then $B(T_iX, Y) = \bar{B}(\Pi T_iX, \Pi Y) = \bar{B}(\Pi X, \Pi Y) = B(X, Y)$ by (H1), and likewise in the second slot. □

Status: elementary corollary of Theorem 4.1, equivalently the universal property of the coinvariants $V_G \cong O$ applied to bilinear forms (Prop. 6.3(a)). What is new is the explicit separation of the hypotheses, below.

Corollary 7.2 (Hierarchy)**.** *Under (H1) the following implications hold and none can be reversed in general:*

$$\textit{independent invariance} \implies \textit{joint invariance}, \qquad \textit{factoring through } \Pi \implies \textit{independent invariance}.$$

Under (H1)–(H2) additionally independent invariance $\implies$ factoring through Π. Joint invariance $\not\implies$ factoring, even under (H1)–(H2) (hidden-sector model).

7.3 Sharpness of the hypotheses

Each hypothesis of Theorem 7.1 is necessary in the following sense.

Example 7.3 (Independent invariance without (H2))**.** $V = \mathbb{R}^2$, $\Pi(x, y) = x$, $\mathcal{T} = \{\mathrm{id}\}$, $B(X, Y) = yy'$. (H1) holds, B is trivially independently invariant under id, but $\mathrm{span}\{X - X\} = 0 \subsetneq K = \mathrm{span}\{(0, 1)\}$, and B does not descend ($\delta = 1$). Refinements must be *rich enough* to generate the observable-null directions.

Example 7.4 (One-sided invariance)**.** $V = \mathbb{R}^2$, $\Pi(x, y) = x$, $T(x, y) = (x, -y)$ (so (H1), (H2) hold) and $B(X, Y) = x\,y'$ (Example 4.4). B is left-invariant, $B(TX, Y) = B(X, Y)$, but not right-invariant, and does not descend. *Both* slots must be refined independently.

Example 7.5 (Joint invariance only)**.** The hidden-sector model (Prop. 5.1) satisfies (H1), (H2), joint invariance and fails independent invariance and descent.

Example 7.6 (Permutation refinements and a sum observable)**.** Let $V = \mathbb{R}^n$ ($n \geq 2$), $\Pi(x) = \sum_i x_i \in \mathbb{R}$, and let $\mathcal{T} = \{T_i\}$ be the adjacent transpositions (i $i{+}1$), acting by permuting coordinates. (H1) holds. $X - T_iX = (x_i - x_{i+1})(e_i - e_{i+1})$, so $\mathrm{span}\{X - T_iX\} = \mathrm{span}\{e_i - e_{i+1}\} = \{\sum x_i = 0\} = \ker \Pi$: (H2) holds. The space of bilinear forms invariant under all T_i *jointly* is two dimensional, spanned by $\langle x, y\rangle = \sum x_i y_i$ and $(\sum x_i)(\sum y_i)$; the space of

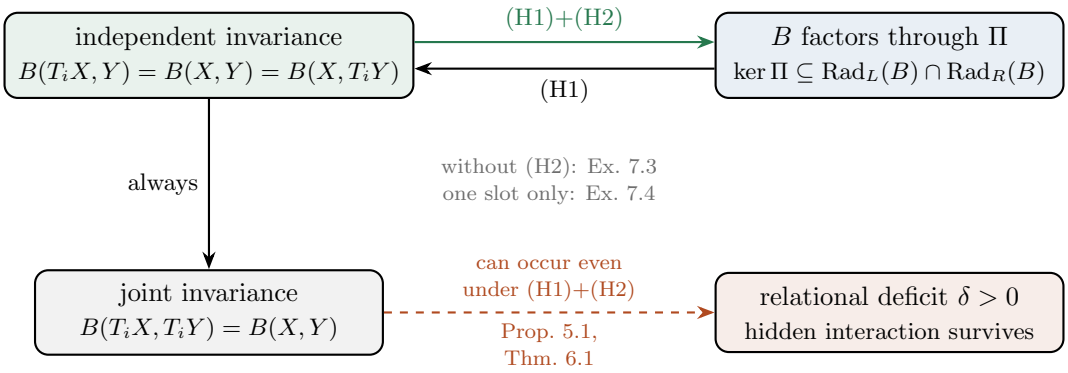


Figure 4: The refinement-compatibility hierarchy. Independent invariance is the only refinement condition that reaches factorisation, and only together with (H1) and the kernel-generation hypothesis (H2). Joint invariance, however strong the symmetry, stops short: the converse of the left-hand vertical arrow fails.

independently invariant forms is one dimensional, spanned by $(\sum x_i)(\sum y_i)$ — exactly the forms that factor through Π, as Theorem 7.1 predicts. For $B = a\langle x, y\rangle + b(\sum x)(\sum y)$ with $a \neq 0$, $K \cap \mathrm{Rad}(B) = 0$ and $\delta = n - 1$.

Verification. Joint invariance under all transpositions makes B an S_n-equivariant endomorphism of the permutation module $\mathbb{R}^n = \mathbf{1} \oplus \{\sum = 0\}$, which has exactly two irreducible summands (non-isomorphic for $n \geq 2$) — a two-dimensional commutant. Independent invariance forces the $\{\sum = 0\}$ block to vanish by Theorem 7.1 (equivalently, $B(e_i - e_{i+1}, \cdot) = 0$ for all i). The dimension counts are reproduced numerically for $n = 5$ in Appendix B. □

The example shows the theorem at work in a coarse-graining setting: a macroscopic sum is a complete observable for interactions that are invariant under *each* micro-relabelling separately, and incomplete for interactions that are invariant only under simultaneous relabelling.

7.4 The compatibility hierarchy

Figure 4 summarises Theorem 7.1 and Corollary 7.2, together with the examples that block each reverse arrow.

7.5 Quantitative form

Proposition 7.7. *Assume (H1)–(H2) with V of finite dimension. For any bilinear B, $\delta(\Pi, B) = \dim q\big(\mathrm{span}\{X - T_iX\}\big)$, where $q\colon V \to V/\mathrm{Rad}(B)$ is the quotient map; i.e. the deficit is the dimension of the image of the refinement differences in the interaction quotient. In particular $\delta = 0$ iff B is independently refinement invariant.*

Proof. $\delta = \dim K/(K \cap \mathrm{Rad}) = \dim q(K)$, and $K = \mathrm{span}\{X - T_iX\}$ by (H2). The last statement is Theorem 7.1. □

8 Generative Response Profiles

Sections 3–7 ask what one projection Π and one pairing B can and cannot say about a route. A generative description should say more: it should show how generative constraints become observable *consequences*, rather than assert that a structure is present. We therefore add one ingredient, a transformation operator, and ask which distinctions between routes survive in the way a route *responds*. Everything in this section is a proposed formal extension of the model of §2; none of it is a canonical UNNS definition. The one statement of a programmatic kind (Principle 8.14) is labelled as a proposal and kept apart from the propositions, which are proved.

8.1 Mechanisms, realisations and response profiles

Definition 8.1 (Mechanism and realisation)**.** A *mechanism* is a quadruple $M = (\Sigma, R, C, O)$ of ingredients, generation rules, admissibility constraints and ordering, in the vocabulary of [15]. A *realisation* of M on a route space (V, Π) is a pair (B, F) of a bilinear pairing B and a linear *transformation operator* $F\colon V \to V$ (a refinement step, an evolution step or a perturbation). The assignment $M \mapsto (B, F)$ is part of the model and is not derived here. F need not preserve Π; if $\Pi F = \Pi$ it is an observable-preserving refinement in the sense of Definition 2.6.

Definition 8.2 (Generative response profile)**.** For $X \in V$ the *generative response profile* is

$$G(X) = \Big(\big(\Pi F^n X\big)_{n\geq 0},\ \big(B(F^n X, \eta)\big)_{n\geq 0,\ \eta\in V} \Big). \tag{12}$$

The first component records what is present and how it persists under transformation; the second records how the transformed route interacts with every probe η. Write $X \sim_G X'$ (*profile equivalence*) if $G(X) = G(X')$.

The profile uses the left slot of B. For symmetric B nothing is lost; the right-slot version exchanges Rad_L and Rad_R below.

Proposition 8.3 (Profile congruence)**.** *G is linear, so $X \sim_G X'$ iff $X - X' \in S_G$, where*

$$S_G = \bigcap_{n\geq 0} F^{-n}\big(K \cap \mathrm{Rad}_L(B)\big) \tag{13}$$

is the largest F-invariant subspace contained in $K \cap \mathrm{Rad}_L(B)$. If $\dim V = d < \infty$ the intersection may be truncated at $n \leq d-1$.

Proof. $G(Z) = 0$ iff $\Pi F^n Z = 0$ and $B(F^n Z, \cdot) = 0$ for all n, i.e. $F^n Z \in W := K \cap \mathrm{Rad}_L(B)$ for all $n \geq 0$; this is (13). The set so defined is F-invariant and lies in W; and every F-invariant subspace of W consists of vectors whose whole F-orbit stays in W, hence lies in it. The truncation is the classical rank argument of Kalman [6, 11]. □

In the language of linear systems, S_G is the *unobservable subspace* of the system with transition F and output $Z \mapsto (\Pi Z, B(Z, \cdot))$. The proposition is therefore standard; what is new here is its reading in the route-system setting.

Corollary 8.4 (Levels of identity)**.** *(a)* $0 \subseteq S_G \subseteq K \cap \mathrm{Rad}_L(B) \subseteq K$.

So route identity implies profile equivalence, which implies observable equivalence and equality of left interaction classes, $B(X, \cdot) = B(X', \cdot)$.

(b) Observable equivalence implies profile equivalence iff $FK \subseteq K$ and $K \subseteq \mathrm{Rad}_L(B)$.

(c) If $\Pi F = \Pi$, then $FK \subseteq K$ automatically and (b) reduces to $K \subseteq \mathrm{Rad}_L(B)$, the left half of Theorem 4.1. Moreover $\Pi F^n X = \Pi X$ for all n: the transformation channel is silent, and every distinction inside a fibre is carried by B.

(d) $S_G = 0$ iff G is injective, i.e. the profile determines the route.

Proof. (a) is read off (13). (b): $K \subseteq S_G$ forces $K \subseteq K \cap \mathrm{Rad}_L(B)$ and $FK \subseteq FS_G \subseteq S_G \subseteq K$; conversely an F-invariant $K \subseteq \mathrm{Rad}_L(B)$ is an F-invariant subspace of $K \cap \mathrm{Rad}_L(B)$, hence lies in S_G. (c) $\Pi F Z = \Pi Z = 0$ for $Z \in K$. (d) G is linear. □

Level	Question	Formal object
Observable identity	what is seen?	$X - X' \in K$
Relational (response) identity	how does it transform, persist and interact?	$X - X' \in S_G$
Generative identity	which admissible mechanism produces it?	route level: $X = X'$ in V; mechanism level: the class of M modulo $\sim_P$ (Def. 8.7)

The three levels need not coincide. The route-level inclusions of Corollary 8.4(a) are the response-layer form of the three equivalences of Definition 3.4, and the following examples show that each can be strict.

Example 8.5 (Same observable, same refinement, opposite verdicts)**.** In the coordinates $X = (u, v) \in \mathbb{R}^3 \oplus \mathbb{R}^3$ of Proposition 5.1 take $\Pi(u, v) = u + v$ and $F = T$, $T(u, v) = (v, u)$. Compare $B_H = -(u \cdot a + v \cdot b)$ (the hidden-sector pairing) with $B_1 = -\Pi^\top \Pi$, which factors through Π. For $X = (e_x, 0)$, $X' = (0, e_x)$ and $Y = X$ we have $\Pi X = \Pi X' = e_x$; $B_H(X, Y) = -1$ but $B_H(X', Y) = 0$, whereas $B_1(X, Y) = B_1(X', Y) = -1$. Since $\mathrm{Rad}_L(B_H) = 0$ we get $S_G = 0$ for B_H, while $\mathrm{Rad}_L(B_1) = K$ gives $S_G = K$ ($\dim 3$) for B_1. The same observable and the same refinement thus carry different generative characters. Since $\Pi T = \Pi$, the whole difference lies in the relational channel (Corollary 8.4(c)); the individual observation does not show it.

Example 8.6 (A distinction invisible to Π and B that the dynamics reveals)**.** Let $V = \mathbb{R}^2$, $\Pi(x, y) = x$, $B(X, Y) = xx'$. Then $K = \mathrm{span}\{e_y\} = \mathrm{Rad}(B)$, so B factors through Π, $\delta = 0$, and at $n = 0$ the profiles of e_y and 0 coincide. Take two realisations with the same B: M_1 with $F_1 = \mathrm{id}$ and M_2 with the shear $F_2(x, y) = (x + y, y)$. Then $S_G(M_1) = K$ but $S_G(M_2) = 0$, because $\Pi F_2^n X = x + ny$. The two realisations agree on every output of order $n = 0$ and are separated by the first transformation response ΠF; moreover $\Pi F_2^n X$ stays bounded iff $y = 0$, a persistence property that M_1 lacks.

8.2 Forward and inverse directions

A mechanism generates routes and their responses (forward); an observer holds only recorded outputs and asks which routes and mechanisms are compatible with them (inverse). Figure 5 separates the two. Proposition 8.3 describes what the inverse direction can resolve at the level of routes; the next two statements describe what it cannot resolve at the level of mechanisms.

Definition 8.7 (Observation protocol; observational equivalence of mechanisms)**.** An *observation protocol* $P = (N, \mathcal{P}, A)$ consists of a set $N \subseteq \mathbb{N}$ of iterates, a set $\mathcal{P} \subseteq V$ of probes, and a set $A \subseteq V$ of admissible routes. Its output on a realisation (B, F) is $G_P(X) =$

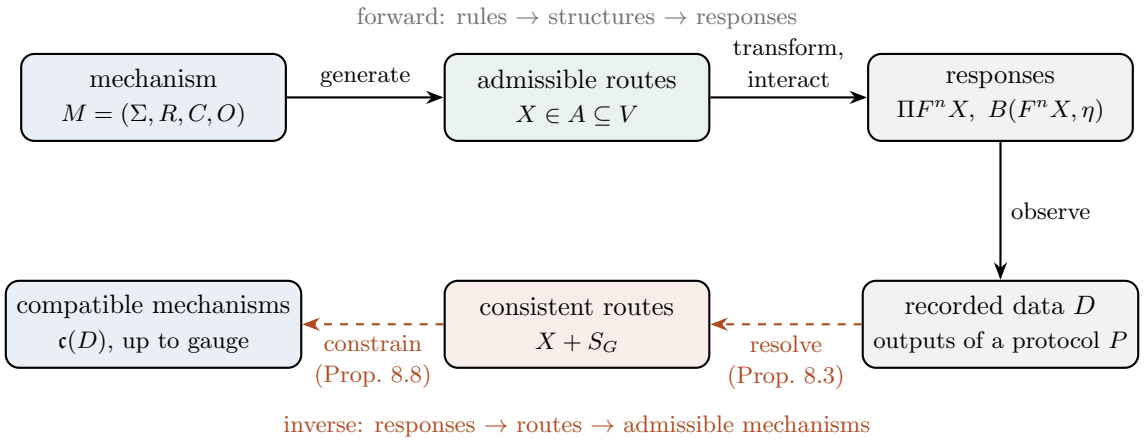


Figure 5: Forward and inverse directions. The forward direction generates structures and responses from rules. The inverse direction can recover a route only up to S_G (Proposition 8.3) and a mechanism only up to the gauge of Proposition 8.8; what it returns is a *class* of compatible mechanisms.

$\big((\Pi F^n X)_{n\in N}, (B(F^n X, \eta))_{n\in N, \eta\in\mathcal{P}}\big)$, $X \in A$. Realisations M_1, M_2 are *P-equivalent*, $M_1 \sim_P M_2$, if $G_P^{M_1}(X) = G_P^{M_2}(X)$ for all $X \in A$; if this holds for every protocol declared admissible we write $M_1 \sim_{\text{obs}} M_2$.

Enlarging N, $\mathcal{P}$ or A can only refine $\sim_P$. If $M_1 \sim_{\text{obs}} M_2$, no evidence from the declared protocols can separate them, and the most that can be claimed is membership of a class.

Proposition 8.8 (Mechanisms are identifiable only up to gauge)**.** *Let* $\dim V < \infty$ *and let* $g \in GL(V)$ *satisfy* $\Pi g = \Pi$. *For a realisation* $M = (B, F)$ *put* $g \cdot M = (B^g, F^g)$ *with* $B^g(X, Y) = B(g^{-1}X, g^{-1}Y)$ *and* $F^g = gFg^{-1}$. *Then:*

(a) $\Pi(F^g)^n(gX) = \Pi F^n X$ *and* $B^g\big((F^g)^n(gX), g\eta\big) = B(F^n X, \eta)$ *for all* $X, \eta \in V$ *and* $n \geq 0$. *Thus M and $g\cdot M$ produce the same set of responses* when routes and probes are relabelled by g together.

(b) $\dim K$, ρ_L, ρ_R, δ *and* $\dim S_G$ *take the same values on M and $g \cdot M$.*

(c) If g fixes every route of A and every probe of $\mathcal{P}$, then $M \sim_P g \cdot M$ for $P = (N, \mathcal{P}, A)$. Without that condition the fixed-label outputs of M and $g \cdot M$ may differ.

The group $\{g : \Pi g = \Pi\}$ *contains every invertible* $\mathrm{id} + N$ *with* $N \in \mathrm{Hom}(V, K)$, *so it is non-trivial whenever* $K \neq 0$.

Proof. (a) $\Pi g = \Pi$ gives $\Pi g^{-1} = \Pi$, and $(F^g)^n = gF^n g^{-1}$; then $\Pi(F^g)^n gX = \Pi g F^n X = \Pi F^n X$ and $B^g(gF^n X, g\eta) = B(F^n X, \eta)$. (b) $\Pi g = \Pi$ implies $gK \subseteq K$, hence $gK = K$. Also $\mathrm{Rad}_L(B^g) = g\,\mathrm{Rad}_L(B)$ and $\mathrm{Rad}_R(B^g) = g\,\mathrm{Rad}_R(B)$, so $K \cap \mathrm{Rad}_{L/R}(B^g) = g\big(K \cap \mathrm{Rad}_{L/R}(B)\big)$, and $S_G(g \cdot M) = gS_G(M)$ by conjugacy of F and F^g. All dimensions are preserved. (c) For $X \in A$, $\eta \in \mathcal{P}$: $\Pi(F^g)^n X = \Pi g F^n g^{-1} X = \Pi F^n X$ and $B^g((F^g)^n X, \eta) = B(F^n X, g^{-1}\eta) = B(F^n X, \eta)$. For the last claim, $\Pi(\mathrm{id} + N) = \Pi$. □

Remark 8.9 (Relabelled versus fixed-label equivalence)**.** Part (a) is an identity between M evaluated at (X, η) and $g \cdot M$ evaluated at $(gX, g\eta)$; it is not an identity at fixed labels. Example: $V = \mathbb{R}^2$, $\Pi(x, y) = x$, $B(X, Y) = y_X y_Y$, $F = \mathrm{id}$, $g(x, y) = (x, 2y)$. At the fixed route and probe e_y the output of M is 1 while that of $g \cdot M$ is $B(g^{-1}e_y, g^{-1}e_y) = \frac{1}{4}$; at

(ge_y, ge_y) it is 1 again. Hence the P-equivalence of Definition 8.7, which compares outputs at fixed labels, is finer than equivalence up to relabelling, and the two coincide only for the gauge elements allowed by (c). If an experiment names its routes and probes operationally, only g fixing them remain unidentified; if it does not, the full group acts.

Labels of the unobservable directions of V are therefore identifiable only to the extent that the protocol names routes and probes (Remark 8.9); the *invariants* of the isomorphism class of (V, Π, B, F) over Π are. This is what "relational identity" can mean operationally: a statement about δ, S_G or ρ_L is meaningful, a statement about which element of V a route "really is" is not, unless the protocol names it.

Proposition 8.10 (Probe complexity)**.** *Assume* $\dim K < \infty$ *(e.g.* $\dim V < \infty$*); "generic" means outside a proper algebraic subset of* V^m*. Let* $\rho_L = \dim K/(K \cap \mathrm{Rad}_L(B))$ *and let* $\eta_1, \dots, \eta_m \in V$ *be probes. The routes of the fibre* $X_0 + K$ *that no probe separates from* X_0 *at* $n = 0$ *are* $X_0 + K_J$*,* $K_J = \{Z \in K : B(Z, \eta_j) = 0,\ j \le m\}$*. Then* $\dim K_J \ge \dim K - m$*, and for generic probes* $\dim K_J = \dim K - \min(m, \rho_L)$*. In particular* ρ_L *generic probes reduce the ambiguity to* $K \cap \mathrm{Rad}_L(B)$*, which no left probe can resolve, and fewer than* ρ_L *probes cannot.*

Proof. The map $K \to V^*$, $Z \mapsto B(Z, \cdot)$, has kernel $K \cap \mathrm{Rad}_L(B)$ and rank ρ_L; so $\Lambda = \{B(\cdot, \eta)|_K : \eta \in V\} \subseteq K^*$ has dimension ρ_L (the rank of the same matrix, read column-wise). K_J is the annihilator in K of m elements of Λ. These span at most m and, generically, $\min(m, \rho_L)$ dimensions. If $m < \rho_L$ then $\dim K_J \ge \dim K - m > \dim K - \rho_L = \dim(K \cap \mathrm{Rad}_L(B))$. □

Since $\rho_L \le \delta$ (Proposition 4.7), this is the experimental counterpart of the statement that δ additional observables restore descent: here ρ_L interaction tests resolve the fibre up to what no test can see.

8.3 Admissibility as restriction

Definition 8.11 (Admissible set and its fibre differences)**.** Let $A \subseteq V$ be the set of routes admitted by the constraints C of a mechanism, and put $K_A = \mathrm{span}\{X - X' : X, X' \in A,\ \Pi X = \Pi X'\} \subseteq K$.

Proposition 8.12 (Restriction)**.** *For* $X, X' \in A$*, the implication* $\Pi X = \Pi X' \Rightarrow B(X, \eta) = B(X', \eta)\ \forall \eta$ *holds iff* $K_A \subseteq \mathrm{Rad}_L(B)$*. The restricted left deficit* $\delta_A = \dim K_A/(K_A \cap \mathrm{Rad}_L(B))$ *satisfies* $\delta_A \le \rho_L$ *and can vanish while* $\delta > 0$*.*

Proof. By linearity of B in the first slot the implication holds for all pairs iff every generating difference lies in $\mathrm{Rad}_L(B)$, i.e. iff $K_A \subseteq \mathrm{Rad}_L(B)$. $K_A/(K_A \cap \mathrm{Rad}_L(B))$ injects into $K/(K \cap \mathrm{Rad}_L(B))$. For the last claim, in Example 8.5 take $A = \{(u, 0)\}$: Π is injective on A, so $K_A = 0$ and $\delta_A = 0$, whereas $\delta(\Pi, B_H) = 3$. □

Restriction is only as informative as the admissibility principle that imposes it is independent of the observations: choosing A to be a section of Π removes every obstruction by fiat.

8.4 What would make a generative description explanatory

Definition 8.13 (Predicted and discriminating responses)**.** Let P_0 be a calibration protocol with recorded outputs D, and let $\mathfrak{c}(D)$ be the class of realisations whose P_0-output equals D. A realisation $M \in \mathfrak{c}(D)$ *predicts* the output $q(M)$ of any protocol $P \supsetneq P_0$. The prediction is *discriminating* if $q(M) \neq q(M')$ for some $M' \in \mathfrak{c}(D)$.

In Example 8.6, M_1 and M_2 lie in the same class for $P_0 = (\{0\}, V, V)$; the response ΠF discriminates them. Four requirements for a generative *explanation* then have formal counterparts, with one that is methodological:

Requirement	Formal counterpart	Status
Construction	(B, F) fixed from $M = (\Sigma, R, C, O)$ without importing the target observable	protocol-level condition (pre-registration); not an algebraic statement
Restriction	admissible set A, fibre differences K_A, restricted deficit δ_A (Prop. 8.12)	formal; its force depends on independent justification of C
Prediction	outputs of $P \supsetneq P_0$ determined by $M \in \mathfrak{c}(D)$ (Def. 8.13)	formal; the prediction is fixed before validation data are inspected
Discrimination	predictions that split $\mathfrak{c}(D)$ (Ex. 8.6)	formal; instance in the abstract model only

8.5 The proposed principle

Principle 8.14 (Generative characterisation; proposed)**.** The generative nature of an observable structure is characterised by the admissible mechanisms capable of producing it, together with the transformation, persistence, boundary and interaction relations those mechanisms imply. Observable identity alone need not determine generative identity; generative explanations become scientifically informative when they constrain or predict observable behaviour beyond the data used to construct them.

This is a proposed foundational principle, *not a theorem*. Its mathematical content in the present model is the following dictionary; the empirical content, whether canonical UNNS recursion supplies independently justified admissibility principles that make such predictions, is not established.

Clause of the principle	Formal object here	Status
admissible mechanisms capable of producing it	compatible class $\mathfrak{c}(D)$; restriction through A, K_A	definitions; whether UNNS rules fix A is open
transformation, persistence	F; $\Pi F^n X$; S_G (Ex. 8.6)	defined; example only
interaction	$B(F^n X, \eta)$; ρ_L, δ	defined and proved
boundary	not modelled (linear F has no exit from A)	**open** (L7)
observable identity does not determine generative identity	Thm. 4.1; Cor. 8.4; Ex. 8.5	proved
only a class of mechanisms is identifiable	Prop. 8.8	proved
informative beyond calibration data	Def. 8.13	definition; no UNNS instance

Remark 8.15 (Weak and strong claims)**.** *Weak claim:* every observable structure has some generative description. In this framework it is vacuous: the realisation $V = O$, $\Pi = \mathrm{id}$ has $K = 0$, $\delta = 0$ and $S_G = 0$ for any B, F. *Strong claim:* a mechanism whose admissibility principles are justified independently of the observations restricts the admissible routes (small K_A) and predicts discriminating responses. Nothing in this paper establishes the strong claim for canonical UNNS; the machinery above only makes it precise enough to be tested, and Problem 11.4 records what a test would need.

9 Reproducible Constructions

Every example above is finite-dimensional or reduces to a finite computation, and has been checked by a single deterministic script, `verify_ri.py`, delivered with this manuscript. This section records the design of the checks and their outcome; the full output is reproduced in Appendix B.

9.1 Design

- **Exact arithmetic where the statement is exact.** The Legendre witnesses (Example 5.4, Table 3) are computed with rational arithmetic (`fractions.Fraction`), and the four-route table (Example 5.5) in $\mathbb{Q}(\sqrt{3})$ by a small exact class; the identities $P_3(\frac{1}{2}) = -\frac{7}{16}$, $B_3(X, r(\mathbf{e}_x)) = \frac{15}{16}$, $B_5(X, r(\mathbf{e}_x)) = \frac{105}{256}$ are tested for equality, not for tolerance.
- **Seeded randomised tests for the general theorems.** Theorem 4.1, Proposition 4.7, Proposition 6.2, and Theorem 7.1 are tested on random finite-dimensional instances (NumPy generator seeded with a fixed integer), in both "factoring" and "generic" regimes, with the criterion $K \subseteq \mathrm{Rad}_L \cap \mathrm{Rad}_R$ compared against a direct least-squares test for $B = \mathsf{P}^\top \bar{\mathsf{B}} \mathsf{P}$.
- **Response layer.** Proposition 8.3 (the congruence subspace against a direct unobservability computation, and the completeness criterion of Corollary 8.4(b)), Proposition 8.8 (gauge invariance of every protocol output and of $\dim S_G, \rho_L, \delta$), Proposition 8.10 (probe counts) and Proposition 8.12 (restriction) are tested on seeded random instances; Examples 8.5 and 8.6 are checked directly.
- **Rank computation for the obstruction.** For $N = 120$ generic points of S^2 (fixed seed) the matrix of B_ℓ restricted to $\ker \Pi \times V$ is formed and its numerical rank (tolerance 10^{-8}) is compared with $2\ell + 1$, $\ell = 1, \ldots, 7$.
- **Quarantine of the scope.** No check uses a probability, a quantum state or a Bell inequality, apart from the arithmetic $\frac{1}{4}(1 + C)$ in Table 3.

9.2 Checks and outcomes

9.3 What was not re-run

The GCT repository contains its own regression suites (for instance `tests/test_gct_joint_origin_v002.py`). Those were not executed as part of this work and are not relied upon. The route-test analysis that first produced the Legendre counterexample and the (s, h) decomposition is available to us as a transcript [21]; the package of tests that it reports for its own two-state recurrent realisation was *not* among the files provided, so no claim about that

Group	Statement checked	Outcome
H	Prop. 5.1(a)–(f): $B = -(uu' + vv')$, $\Pi T = \Pi$, $h_{TX} = -h_X$, joint invariance, failure of independent invariance, $\operatorname{rank}\operatorname{span}\{X - TX\} = 3$, witness $B(k, Y) = -2$	all pass (max float error $< 2 \times 10^{-15}$)
Q	Thm. 4.1: criterion $\Leftrightarrow$ direct factorisation, 300 random instances; one-sided counterexample	150 factoring, 150 non-factoring; all agree
D	Prop. 4.7: bounds, matrix formula, minimal completion with δ rows (and failure with $\delta - 1$)	300 random cases pass
L	Thm. 5.3: exact witnesses $\ell = 1, 3, 5$; ranks for $\ell \leq 7$; positivity of $-B_\ell$; odd-ℓ sign reversal	ranks $0, 5, 7, 9, 11, 13, 15$ $= 2\ell + 1$ for $\ell \geq 2$; $-B_\ell \succeq 0$
I	Prop. 6.2 and Thm. 7.1: 200 random involutions, block structure and compatibility; coinvariant gap $\dim V_-^{\otimes 2} = 9$	all pass
P	Example 7.6, $n = 5$: dimension of joint-invariant forms = 2, independent-invariant forms = 1; (H2); $\delta = 4$	2, 1; pass; 4
C	§10.4: ideal Helmholtz projector per Fourier mode ($\mathbf{k} = (1, 2)$)	idempotent; R involution; joint but not independent invariance; $\delta = 1$
G	Remark 5.7: $c_3(d) = -P_3(\cos d)$ satisfies $c(0) = -1$, evenness, $c(d + \pi) = -c(d)$, $\lvert c\rvert \leq 1$	pass (41 sample angles)
F	Example 5.5: Gram matrices of B_1, B_3 on four routes; $G_1 k = 0$, $G_3 k = (\frac{15}{16}, \frac{5\sqrt{3}}{16}, 0, -\frac{15}{16})^\top$	exact over $\mathbb{Q}(\sqrt{3})$; pass
S	Prop. 4.8: Q descends iff B descends (100 random symmetric forms); Prop. 4.9: induced pairing $-\frac{1}{2} s^\top (\mathsf{I} + H^\top H) s'$	pass
R	§8: Props. 8.3, 8.8, 8.10, 8.12 (300/300/200/200 random instances); Ex. 8.5 witness $B(X, Y) = -1$, $B(X', Y) = 0$, S_G of dimensions 0 and 3; Ex. 8.6 (shear separates, identity does not); trivial realisation $V = O$	all pass
N	Prop. 10.1: identity $M \perp N = M(N + 1)^2/N$ (exact, 59×59); fibre of 72 is $\{(18, 1), (16, 2), (10, 5)\}$; 235 multi-route values for $M \leq 200$, $N \leq 40$	all pass (exact rational arithmetic)

Table 4: Summary of the verification. All 58 checks pass; the script exits with status 0. Raw output: Appendix B.

realisation is verified here. The models of this paper are the abstract algebraic ones defined above, and the verification concerns them only.

10 Implications for UNNS

This section relates the results to canonical UNNS. It keeps three kinds of statement apart: a *proven consequence* (a statement about canonical UNNS objects that follows by proof from adopted definitions alone), a *conditional consequence* (valid for any interaction or transformation that is specified independently and satisfies a stated hypothesis), and a *proposed extension* (a definition or principle introduced here). It asserts no property of canonical UNNS recursion beyond the first kind.

10.1 Canonical UNNS: structural commitments and mathematical consequences

Not having proved a theorem about the interaction of a canonical mechanism does not make the results irrelevant to UNNS. The entry grammar commits UNNS to a particular reading of a number: a value is structurally meaningful as a *generated occurrence*, carrying a rule, a position, a route, a boundary and a status, so that two occurrences of the same numeral may differ in status [22]. The question treated here, what is lost when only the value is retained, is therefore a question about canonical UNNS. The correspondence below states which features are inherited, which are formal representations, and which are new.

Canonical UNNS concept	Object in this manuscript	Origin
Generated occurrence (M, N) with rule, route, boundary, status	elementary route $r(M, N)$; route space $V = F(E)$	concept inherited; the free linear span is a *formal representation* introduced here
Generated value $M \perp N$	observation Π	value inherited; linear extension formal
Same occurrence (same rule, position, route)	generative identity, equality in V	inherited on E; canonicalisation relations for combinations: new, unspecified
Admissibility, five criteria	admissible set A, fibre differences K_A	criteria inherited; their use as restriction is new
Recursive operation	transformation operator F	concept inherited; a linear F for the nest rule is not specified (open)
Structure-preserving morphism	refinement family $\mathcal{T}$, $\Pi T = \Pi$	concept inherited; identification with Π-preserving maps new
Generative interaction	pairing B	**not defined in the entry grammar [22]**; must be specified independently (open)
Lost relational distinction	$K/(K \cap \operatorname{Rad} B)$	formal; conditional on B

Proposition 10.1 (Value does not determine route; proven consequence)**.** *The nest operation satisfies* $M \perp N = MN + M/N + 2M = M(N+1)^2/N$. *Among positive integer pairs, the value* 72 *is generated by exactly three occurrences,* $(M, N) = (18, 1), (16, 2), (10, 5)$. *Each lies on an integer-anchor route (*$M = kN$ *with* $k = 18, 8, 2$*), for three different nests and three different boundary conditions* $N \mid M$.

Proof. The identity is algebra. For 72 we need $M = 72N/(N+1)^2 \geq 1$, i.e. $N \leq 69$; checking $N \leq 69$ leaves exactly $N = 1, 2, 5$ with $M = 18, 16, 10$, and $k(N+1)^2 = 72$ in each case. (Exact enumeration: Appendix B.) □

Thus the observable value 72 does not determine the generated occurrence, which is the entry grammar's own point about numerals [22]; in the linear representation, $r(10, 5) - r(18, 1)$ and $r(10, 5) - r(16, 2)$ lie in $K = \ker \Pi$. This is a statement about canonical objects (the set-level non-injectivity of the value map) and needs no linear model. By direct enumeration 235 values with $M \leq 200$, $N \leq 40$ have at least two occurrences.

Conditional consequence. For *any* bilinear interaction B on V that is specified independently, Theorem 4.1 says that the interaction of occurrences is determined by their values iff differences of colliding occurrences lie in $\operatorname{Rad}(B)$; a pairing that separates $(18, 1)$ from $(10, 5)$ against some probe cannot descend to the value. The entry grammar does not define such a

pairing, so whether canonical UNNS retains interaction information invisible to the value is **not demonstrated**.

The bridge. The research obligation is a map

$$\boxed{\text{canonical mechanism} \longrightarrow (V, \Pi, B, \mathcal{T}, F)}$$

constructed for at least one authentic mechanism, with B determined independently of the desired theorem. For the nest rule, the pair (V, Π) is supplied by Proposition 10.1; F, $\mathcal{T}$ and B are not. If the bridge is built, the manuscript can make a direct claim about that mechanism; if not, the results remain model-conditional (Problem 11.3).

Principle 10.2 (Scoped statement for the UNNS Substrate; proposed)**.** In the UNNS Substrate, the characterisation of generated structures includes their routes, admissibility conditions and transformation histories. Consequently, equality of observable representations alone does not establish equality of generative descriptions. The present work gives a rigorous algebraic criterion for when observational reduction preserves interaction equivalence and when relational distinctions may survive it. Its applicability to a particular canonical mechanism depends on constructing the observation and interaction maps from that mechanism's independently specified rules.

The first two sentences rest on the adopted grammar and Proposition 10.1; the third is Theorem 4.1; the fourth is the conditional consequence above.

10.2 Structural identity

The three equivalences of Definition 3.4 (and, with a transformation operator, the profile equivalence of Definition 8.2 between the generative and the observable one) correspond to levels at which a UNNS-style description can speak about a generated occurrence:

Equivalence	Meaning in the route language	Substitution requires
Generative	same structural identity under declared canonicalisation	nothing (strongest)
Observable	same projected observable	$K \subseteq \mathrm{Rad}(B)$ to replace interaction equivalence
Interaction	identical pairings with all other routes	$\mathrm{Rad}(B) \subseteq K$ to replace observable equivalence

The point is not that a particular substitution is false but that *none is automatic*. A description that treats "same observable" as "same structure" is making a compatibility assumption, and Theorem 4.1 states exactly which one.

10.3 Projection and admissibility

A projection Π is part of the specification of an observable realisation. Two disciplines follow.

(D1) *Declare the observation map together with each pairing.* For every interaction B used, either prove $\ker \Pi \subseteq \mathrm{Rad}(B)$ (observable-null compatibility), or report $\delta(\Pi, B)$.

(D2) *Declare the transformation and the protocol.* A claim about the generative character of a structure names the operator F, the iterates and probes used, and the admissible set A (Definition 8.7); without them only Π and B at $n = 0$ are specified, and mechanisms that differ in F are indistinguishable (Example 8.6).

(D3) *Do not infer compatibility from joint symmetry.* Joint refinement invariance is what an exchange or relabelling symmetry of *both* arguments delivers; Theorem 6.1 shows that it leaves the internal sector's own interaction untouched. Independent invariance under a family satisfying (H2) is the condition that does the work (Theorem 7.1).

In the language of the GCT analysis, the *observable-null refinement condition* — a route combination with no net observable cannot contribute to joint pairings — was identified as an additional principle not forced by the route and admissibility rules examined there [21]. In the present formulation it is precisely $K \subseteq \mathrm{Rad}(B)$, equivalently (under (H2)) independent refinement invariance. Whether canonical UNNS admissibility entails it remains open (Problem 11.1); this paper supplies the exact form of the question and examples showing that it is not vacuous.

10.4 Illustration: Operator XVI closure as an observation-like projection

Chamber XVI states the Closure hypothesis as a Helmholtz decomposition that seals flux divergence in an evolved τ-field [19]. We use only the *ideal* L^2 projector on a periodic domain, not the Chamber's discrete implementation (whose source audit records a hybrid discrete/spectral symbol and a closure diagnostic that is not a direct projector-idempotence test [17]). Per nonzero Fourier wavevector $\mathbf{k} \in \mathbb{R}^2$, the solenoidal projector is $\mathsf{P}_\mathbf{k} = \mathsf{I} - \mathbf{k}\mathbf{k}^\top/|\mathbf{k}|^2$ on the amplitude space $V_\mathbf{k} = \mathbb{R}^2$; it is idempotent with kernel $K = \mathrm{span}\{\mathbf{k}\}$ (longitudinal, i.e. gradient, amplitudes). Take $\Pi = \mathsf{P}_\mathbf{k}$ as the "observable" map, the involution $\mathsf{R} = 2\mathsf{P}_\mathbf{k} - \mathsf{I}$ (which reverses the longitudinal part), and the L^2 pairing $B(X, Y) = X \cdot Y$. Then R preserves Π, $B(\mathsf{R}X, \mathsf{R}Y) = B(X, Y)$, $B(\mathsf{R}X, Y) \neq B(X, Y)$ in general, $\mathrm{Rad}(B) = 0$ and $\delta = 1$. This is the structure of Proposition 6.2 with V_+ the solenoidal and V_- the longitudinal part. In words: the L^2 pairing of two fields is invariant under reflecting the compressive component of *both*, but a pairing that depended on the closed field alone would have to ignore the compressive component entirely. This is a formal illustration of the abstract structure; it makes no statement about the numerical behaviour of Chamber XVI.

10.5 Relational information

Define the *relational residue* of (Π, B) as the quotient $\mathcal{R}(\Pi, B) = K/(K \cap \mathrm{Rad}\, B)$, with $\dim \mathcal{R} = \delta$. It is the part of the generative description that is invisible to Π but visible to B; in the algebraic sense used here, "relational information" that is lost by observation is exactly this residue. We use the word information only in this algebraic sense; no entropy or probability is defined. The compatibility hierarchy (Figure 4) says that under (H1)–(H2) the residue vanishes if and only if the interaction is independently refinement invariant.

10.6 Possible applications (not validated)

Questions of the same form — does an interaction between microscopic configurations depend on them only through a macroscopic variable? — occur in coarse-grained dynamical systems, multiscale modelling, structural representations and accounts of emergent observables. Example 7.6 is the prototype: a sum observable is complete for interactions invariant under each relabelling separately and incomplete otherwise. We offer the criteria of Theorems 4.1 and 7.1 as candidate tests in such settings, and present these as *possible* applications. No empirical claim is made, and no physical system is analysed.

11 Discussion and Limitations

11.1 Relation to existing theory

Quotient spaces and bilinear algebra. Theorem 4.1 is the universal property of quotients for bilinear maps; radicals of bilinear forms and the nondegenerate quotient pairing are classical [8, 10, 5]. Theorem 7.1 is the universal property of coinvariants [3] applied to bilinear forms, plus the observation that coinvariant-generation (H2) is a hypothesis on the observation map rather than on the interaction. We claim no novelty of technique.

Coarse-graining. Descent criteria of this type recur in coarse-graining. For linear dynamics, strong lumpability of a Markov chain with respect to a partition [7] is the statement that the aggregation map Π on measures satisfies $A(\ker \Pi) \subseteq \ker \Pi$ for the transition operator A; for bilinear pairings the analogous descent condition is $\ker \Pi \subseteq \mathrm{Rad}(B)$. The two conditions are of the same kind (a structure descends to the quotient iff it respects the kernel) but are not equivalent and should not be conflated.

Linear systems. Proposition 8.3 is the characterisation of the unobservable subspace of a linear system [6, 11]; the response profile is its input-free output sequence augmented by an interaction channel. We claim no novelty for the proposition, only for its use as the relational layer of a route system.

Hidden-state models. The decomposition $V = O \oplus H$ is formally a hidden-state structure. We use it only as algebra: there is no probability measure on H, no dynamics, no notion of locality, and nothing here constrains correlations. In particular nothing in this paper bears on Bell-type inequalities [2, 4, 12], and any account of experimental correlations built on relational invariants would have to satisfy no-signalling and the established Bell constraints independently.

11.2 Limitations

(L1) All structures are linear and all interactions bilinear. Nonlinear route composition, or interactions of higher order (trilinear, tensor), are outside the scope.

(L2) The field is $\mathbb{R}$ throughout; the characteristic-$\neq 2$ hypothesis is used in Propositions 6.2 and 6.3.

(L3) The analysis is algebraic. In infinite dimensions no topology or continuity is imposed, so "kernel" and "radical" are algebraic subspaces; a Hilbert-space version would require closed kernels and bounded forms.

(L4) The route spaces are *constructed*; they are not derived from canonical UNNS grammar. All hypotheses (linearity, bilinearity, the form of Π, the refinements) are hypotheses of a proposed model.

(L5) The Legendre family is a specific example family with $O(3)$ covariance; the obstruction rank $2\ell + 1$ is a property of that family, not a general law.

(L6) The verification (§9) concerns the abstract models; it does not test any UNNS engine.

(L7) The transformation F is linear and the admissible set A is a subset of a vector space. Boundary behaviour, where an orbit leaves an admissible region and the regime changes, requires nonlinear admissibility and is not modelled; the "boundary" clause of Principle 8.14 is therefore not captured.

(L8) The assignment from a mechanism (Σ, R, C, O) to a realisation (B, F) is assumed, not derived; the profile uses the left slot of B only.

11.3 What is and is not established

Statement	Status
$\Pi X = \Pi X' \not\Rightarrow B(X,Y) = B(X',Y)$ in the models above	proved (Thm. 6.1, Thm. 5.3), machine-checked
Exact descent criterion $K \subseteq \mathrm{Rad}_L \cap \mathrm{Rad}_R$	proved (standard)
Joint invariance does not imply independent invariance or descent	proved by explicit model
Independent invariance $\Leftrightarrow$ descent under (H1)–(H2)	proved
Canonical UNNS recursion satisfies (H2) with independent refinement invariance	**open**
Canonical UNNS admissibility entails $K \subseteq \mathrm{Rad}(B)$	**open**
The nest value $M \perp N$ does not determine the occurrence (value 72: three routes)	proved (Prop. 10.1); canonical
Canonical UNNS retains interaction information invisible to the value	**not demonstrated** (needs an independent B)
Profile equivalence is $X - X' \in S_G$, between route and observable identity	proved (standard: unobservable subspace)
Mechanisms are identifiable only up to a gauge; $\delta, \rho_L, \dim S_G$ are invariants	proved (Prop. 8.8)
UNNS supplies independently justified admissibility principles that restrict routes and predict discriminating responses (the strong claim)	**open; not established**
Any physical system realises the structure	**not claimed**
Any consequence for quantum correlations	**not claimed**

11.4 Open problems

Open problem 11.1. Determine whether the independently defined route-composition and admissibility rules of canonical UNNS entail $\ker \Pi \subseteq \mathrm{Rad}(B)$ for the interactions they define, or whether they admit a model of the type of Proposition 5.1.

Open problem 11.2. Identify natural refinement families $\{T_i\}$ in UNNS-style grammars for which (H2) holds, i.e. for which observable identity coincides with identity up to refinement.

Open problem 11.3. Construct the map canonical mechanism $\to (V, \Pi, B, \mathcal{T}, F)$ for at least one authentic UNNS mechanism (for instance the nest rule, for which (V, Π) is given by Proposition 10.1), specifying B and F from the mechanism's own rules, fixed before the descent criterion is applied, and decide whether $\ker \Pi \subseteq \mathrm{Rad}(B)$.

Open problem 11.4. Exhibit an admissibility class A derived from canonical UNNS rules, fixed before inspecting any observation, with $\delta_A = 0$ or small K_A (Proposition 8.12), together with a response not determined by the calibration data that discriminates it from a competing admissible mechanism (Definition 8.13). This is the formal content of the strong claim of Remark 8.15.

Open problem 11.5. Extend the response profile to nonlinear admissible transformations with boundaries, so that the "boundary" clause of Principle 8.14 acquires a formal counterpart, and to both slots of B.

Open problem 11.6. Extend Theorems 4.1–7.1 to multilinear interactions and to topological vector spaces, with a robust version in which the deficit is replaced by a norm-distance from the descended subspace.

Open problem 11.7. Characterise all $O(3)$-covariant, sign-reversing, normalised pairings on $F(S^2)$ (the admissibility profile of §5.2); the family B_ℓ and the positive-definiteness of $-B_\ell$ suggest a Schoenberg-type description by positive-definite zonal kernels, which we have not attempted.

11.5 Conclusion

Observable representation, structural identity and interaction are three distinct equivalences on a generative route space, and each implication between them is an inclusion between subspaces. Under the stated hypotheses the descent of an interaction to an observable quotient is governed by one criterion, $\ker \Pi \subseteq \mathrm{Rad}_L(B) \cap \mathrm{Rad}_R(B)$; refinement invariance reaches that criterion only when the refinements are applied to each argument independently and generate the observable-null directions. Simultaneous refinement invariance does not suffice, and the cost of mistaking it for sufficiency is measured by an explicit, computable deficit. The response layer adds that, once a transformation operator is given, the relational identity of a route is the coset of the largest F-invariant subspace of $\ker \Pi \cap \mathrm{Rad}_L(B)$, that mechanisms are identifiable only up to a gauge, and that a generative description becomes informative only through independently justified restriction and discriminating prediction; the proposed Principle 8.14 is a programme for UNNS, not a result. These are results about a proposed mathematical realisation; whether and where canonical UNNS recursion satisfies the compatibility conditions is the question this paper makes precise rather than answers.

A Detailed Algebra

A.1 Coordinates of the hidden-sector model

The map $(u, v) \mapsto (s, h) = (u + v,\, u - v)$ on $\mathbb{R}^3 \oplus \mathbb{R}^3$ is linear with block matrix $\left(\begin{smallmatrix} \mathsf{I} & \mathsf{I} \\ \mathsf{I} & -\mathsf{I} \end{smallmatrix}\right)$ of determinant $(-2)^3 \neq 0$ (each 2×2 block pair $\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ has determinant -2, and there are three), so it is an isomorphism with inverse $u = \frac{1}{2}(s + h)$, $v = \frac{1}{2}(s - h)$. In these coordinates:

$$\begin{aligned} \Pi(s, h) = s, \qquad K = \{(0, h)\}, \\ T(u, v) = (v, u) \iff s \mapsto v + u = s, \quad h \mapsto v - u = -h, \\ B = -\tfrac{1}{2}\big(s\cdot s' + h\cdot h'\big) = -\big(u\cdot u' + v\cdot v'\big). \end{aligned}$$

The Gram matrix of B in (s, h)-coordinates is $-\frac{1}{2}\mathsf{I}_6$, so B is nondegenerate and negative definite; $\mathrm{Rad}(B) = 0$. The left defect of independent invariance is

$$B(TX, Y) - B(X, Y) = -\tfrac{1}{2}\big(s\cdot s' - h\cdot h'\big) + \tfrac{1}{2}\big(s\cdot s' + h\cdot h'\big) = h\cdot h',$$

which is the bilinear form of rank 3 with null space $\{h = 0\}$ in each slot, matching $\delta = 3$.

A.2 Matrix form of the deficit

Let B be the Gram matrix of B in a basis of V and let the columns of K be a basis of $\ker \Pi$. Then $B(\mathsf{K}c, \cdot)$ has coordinate row $c^\top \mathsf{K}^\top \mathsf{B}$, and $B(\cdot, \mathsf{K}c)$ has coordinate column $\mathsf{B}\mathsf{K}c$. Thus

$\mathsf{K}c \in \mathrm{Rad}_L$ iff $\mathsf{B}^\top \mathsf{K}c = 0$ and $\mathsf{K}c \in \mathrm{Rad}_R$ iff $\mathsf{B}\mathsf{K}c = 0$, whence

$$\delta = \dim K - \dim \ker \begin{pmatrix} \mathsf{B}\mathsf{K} \\ \mathsf{B}^\top \mathsf{K} \end{pmatrix} = \operatorname{rank} \begin{pmatrix} \mathsf{B}\mathsf{K} \\ \mathsf{B}^\top \mathsf{K} \end{pmatrix}.$$

Worked examples. (a) §1.2: $\mathsf{B} = \mathsf{I}_2$, $\mathsf{K} = (0,1)^\top$, stacked matrix $\left(\begin{smallmatrix} 0 & 1 & 0 & 1 \end{smallmatrix}\right)^\top$ of rank 1; $\delta = 1$. (b) Example 4.4: $\mathsf{B} = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $\mathsf{K} = (0,1)^\top$: $\mathsf{B}\mathsf{K} = (1,0)^\top$, $\mathsf{B}^\top \mathsf{K} = (0,0)^\top$; $\rho_R = 1$, $\rho_L = 0$, $\delta = 1$, showing $\max(\rho_L, \rho_R) \le \delta \le \rho_L + \rho_R$ with equality at the lower bound. (c) Permutation module, $\mathsf{B} = a\mathsf{I} + b\mathsf{J}$ (J all ones), K any basis of $\{\sum x_i = 0\}$: $\mathsf{J}\mathsf{K} = 0$, so $\mathsf{B}\mathsf{K} = a\mathsf{K}$ and the stacked matrix has rank $n-1$ if $a \neq 0$.

A.3 Spherical-harmonic facts used in Theorem 5.3

We use three classical facts [1]: (1) the real spherical harmonics $S_{\ell m}$ ($m = 1, \dots, 2\ell+1$) of degree ℓ form an orthonormal basis of the degree-ℓ harmonic polynomials restricted to S^2; (2) harmonics of different degrees are $L^2(S^2)$-orthogonal; (3) the addition theorem (10). The restriction to S^2 of a linear function $\mathbf{a} \mapsto w \cdot \mathbf{a}$ is a degree-1 harmonic. Linear independence of $\{S_{\ell m}\}_m$ as functions on S^2 follows from orthonormality, and gives linear independence of the functionals f_m on $F(S^2)$: if $\sum_m c_m f_m = 0$ then evaluating on every generator $r(\mathbf{a})$ gives $\sum c_m S_{\ell m} \equiv 0$.

A.4 Exact Legendre values

x	$P_1(x)$	$P_3(x) = \frac{1}{2}(5x^3 - 3x)$	$P_5(x) = \frac{1}{8}(63x^5 - 70x^3 + 15x)$
0	0	0	0
$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{7}{16}$	$\frac{23}{256}$
1	1	1	1

For the witness $X = r(\mathbf{n}) - \frac{1}{2} r(\mathbf{e}_x) - \frac{\sqrt{3}}{2} r(\mathbf{e}_y)$ of Example 5.4, since $\mathbf{n} \cdot \mathbf{e}_x = \frac{1}{2}$, $\mathbf{e}_x \cdot \mathbf{e}_x = 1$ and $\mathbf{e}_y \cdot \mathbf{e}_x = 0$,

$$B_\ell\big(X, r(\mathbf{e}_x)\big) = -P_\ell(\tfrac{1}{2}) + \tfrac{1}{2} P_\ell(1) + \tfrac{\sqrt{3}}{2} P_\ell(0) = \tfrac{1}{2} - P_\ell(\tfrac{1}{2}) \quad (\ell \text{ odd}),$$

giving 0, $\frac{15}{16}$, $\frac{105}{256}$ for $\ell = 1, 3, 5$.

A.5 The $\mathbb{Z}_2$ coinvariants

Let $T^2 = \mathrm{id}$ on V (characteristic $\neq 2$), $V_\pm$ its ± 1 eigenspaces. The span of $\{X - TX\}$ is $2V_- = V_-$, so $V_G = V/V_- \cong V_+$. For the diagonal action, $T \otimes T$ acts as $+1$ on $V_+ \otimes V_+$ and $V_- \otimes V_-$ and as -1 on the mixed summands; $\mathrm{span}\{w - (T \otimes T)w\}$ equals $2 \cdot$(mixed summands), so $(V \otimes V)_G \cong V_+ \otimes V_+ \oplus V_- \otimes V_-$. The natural map to $V_G \otimes V_G = V_+ \otimes V_+$ kills $V_- \otimes V_- = N$.

A.6 The permutation example

Let P be a permutation matrix and B the Gram matrix of B in the standard basis. Joint invariance is $\mathsf{P}^\top \mathsf{B} \mathsf{P} = \mathsf{B}$, i.e. $\mathsf{B}\mathsf{P} = \mathsf{P}\mathsf{B}$ since $\mathsf{P}^\top = \mathsf{P}^{-1}$. For S_n acting on $\mathbb{R}^n$ the commutant is spanned by I and J. Independent invariance additionally requires $\mathsf{P}^\top \mathsf{B} = \mathsf{B}$: applied to $\mathsf{B} = a\mathsf{I} + b\mathsf{J}$ with $\mathsf{P} \neq \mathsf{I}$ this reads $a(\mathsf{P}^\top - \mathsf{I}) = 0$, so $a = 0$.

B Computational Tests

The script `verify_ri.py` (Python 3, NumPy; no other dependency) is delivered with the manuscript. It is deterministic: random instances come from a NumPy generator (`default_rng`, seed 20261021) and a fixed seed for the point set on S^2; exact statements use rational arithmetic. Run:

```
python3verify_ri.py
```

It prints one line per check, writes `verify_ri_report.json`, and exits with status 0 iff all 58 checks pass. Output of the run reported here:

```
PASS H.B_equals_minus_uu_plus_vv [max err 1.78e-15]
PASS H.T_preserves_Pi
PASS H.T_reverses_h
PASS H.joint_invariance_B(TX,TY)=B(X,Y)
PASS H.separate_invariance_FAILS [max |B(TX,Y)-B(X,Y)| = 11.194]
PASS H.span{X-TX}_has_dim_3=dim_kerPi [rank 3]
PASS H.span{X-TX}_inside_kerPi
PASS H.witness_Pi(k)=0_but_B(k,Y)=-2 [B=-2.0]
PASS H.obstruction_rank_rho_L=3
PASS ResultI.criterion_matches_direct_factorization(300 random) [150 factoring / 150 non-factoring cases]
PASS ResultI.one_sided_radical_insufficient
PASS ResultIII+Prop(block-diagonal): 200 random involutions
PASS Legendre.P3(1/2)=-7/16 [-7/16]
PASS Legendre.B3(X,r(ex))=15/16 exact [15/16]
PASS Legendre.B1(X,r(ex))=0 exact
PASS Legendre.corr_60deg_B1=-1/2,B3=+7/16
PASS Legendre.P(++)_B1=1/8
PASS Legendre.P(++)_B3=23/64=0.359375
  numeric B_1(X,r(ex)) = 0.000000000000
  numeric B_3(X,r(ex)) = 0.937500000000
PASS Legendre.numeric_matches
PASS Legendre.odd_l_sign_reversal
  l=1: rank(B_l|ker Pi)=0 (-B_l PSD: True)
  l=2: rank(B_l|ker Pi)=5 (-B_l PSD: True)
  l=3: rank(B_l|ker Pi)=7 (-B_l PSD: True)
  l=4: rank(B_l|ker Pi)=9 (-B_l PSD: True)
  l=5: rank(B_l|ker Pi)=11 (-B_l PSD: True)
  l=6: rank(B_l|ker Pi)=13 (-B_l PSD: True)
  l=7: rank(B_l|ker Pi)=15 (-B_l PSD: True)
PASS Legendre.B1_factors(rank 0)
PASS Legendre.B_l_l>=2_do_not_factor
PASS Legendre.obstruction_rank=2l+1 (generic N=120) [{1: 0, 2: 5, 3: 7, 4: 9, 5: 11, 6: 13, 7: 15}]
PASS GCTv002.Thm3_symmetry_conditions_hold_for_c3
PASS Coinvariant.dim_gap=dim(V-)^2 for model H
PASS Fig1.same_observable_s=(1,0,0)
PASS Fig1.B(X,X)=-1,B(TX,X)=0,B(TX,TX)=-1
PASS Minimal2D.values
PASS OneSided2D.K_in_RadL_not_RadR
PASS Legendre.l=5_exact [C=-23/256, B5(X,ex)=105/256, P(++)=233/1024]
PASS Deficit.bounds+formula+minimal_completion (300 random) [300 cases]
PASS Perm.joint_invariant_forms_dim=2,separate=1 [joint 2, separate 1]
PASS Perm.deficit=n-1=4
PASS Perm.H2_span{X-T_iX}=ker(sum) (dim 4)
PASS Closure.ideal_Helmholtz_per_mode: P idem, R involution, joint-inv, not separate
PASS Closure.deficit=1_per_mode
PASS FourRoute.k_in_kerPi
PASS FourRoute.G1*k=0
PASS FourRoute.G3*k=(15/16,5sqrt3/16,0,-15/16) [[(15/16+0*sqrt3), (0+5/16*sqrt3), (0+0*sqrt3), (-15/16+0*sqrt3)]]
PASS FourRoute.G3_entries
PASS FourRoute.deficit_B3=1,B1=0
PASS SelfInteraction.Q_descends_iff_B_descends (100 random symmetric)
PASS Sections.model_H_induced_pairing=-1/2(I+H^T H)
PASS Profile.congruence_subspace=largest_F-invariant_subspace_of_K_cap_RadL (300 random) [300 cases]
PASS Profile.completeness: ~Pi => ~G iff K F-invariant and K in Rad_L (300 random)
PASS Profile.hidden-sector: S_G=0 (route fully recoverable from profile); first-order: S_G=K (dim 3)
PASS Identifiability: rho_L generic probes resolve the fibre up to K cap Rad_L; rho_L-1 do not (200 random)
PASS AdmissibleFibres: adequacy on A iff K_A in Rad_L, with K_A a proper subspace of K (200 random)
PASS Witness.same_Pi_different_B: Pi(X)=Pi(X')=e_x, B(X,Y)=-1, B(X',Y)=0
PASS Example.dynamics_separates: Z in K cap Rad_L; M1 (F=id) cannot separate Z from 0, M2 (F=shear) does at n>=1
PASS Gauge: g with Pi g = Pi relabels routes but preserves every protocol output; dim S_G, rho_L, delta invariant (200 random)
PASS Weak_claim.trivial_realisation: Pi=id has K=0, delta=0, S_G=0 for any B,F
PASS Regression.Prop4.9c: O=0 counterexample (unique section, B not descending, K not in Rad)
PASS Regression.Prop8.8: fixed-label output 1 vs 1/4; relabelled (g e_y, g e_y) output 1
PASS Regression.Prop8.8c: g fixing named routes and probes leaves fixed-label outputs unchanged (200 random)
PASS Canonical.nest_identity: M(x)N = M(N+1)^2/N (59x59 exact)
PASS Canonical.fibre_of_72: exactly (18,1),(16,2),(10,5), all anchors M=kN with k(N+1)^2=72
PASS Canonical.non_injective: 235 values in (M<=200,N<=40) have >=2 routes

ALL PASS
```

B.1 Script identity and listing

The script (467 lines) is delivered as a separate file and listed in full below; its digest ties the output above to the exact code that produced it:

`verify_ri.py` SHA-256 `53ac1a9b52c1b9b5aade97605b54a5636680c0884ba1ed47fcb2dda915ffcbb5`

```
#!/usr/bin/env python3
"""Deterministic verification for 'Observable Equivalence and Relational Invariants'.
Exact rational arithmetic (fractions) for model H and the Legendre counterexample;
seeded floating-point checks (numpy) for the general theorems. Prints a report."""
import json, sys
from fractions import Fraction as Fr
import numpy as np

out = {}
ok_all = True
def check(name, cond, detail=""):
    global ok_all
    ok_all &= bool(cond)
    out[name] = {"pass": bool(cond), "detail": detail}
    print(("PASS " if cond else "FAIL ") + name + ((" [" + detail + "]") if detail else ""))

# ---------- Model H: R = R^3 (+) R^3, X=(u,v) ----------
rng = np.random.default_rng(20261021)
def Pi(X): return X[:3] + X[3:] # s = u+v
def h(X): return X[:3] - X[3:] # h = u-v
def B(X, Y): return -0.5 * (Pi(X) @ Pi(Y) + h(X) @ h(Y))
def T(X): return np.concatenate([X[3:], X[:3]])

# identity B = -(u.u' + v.v')
errs = []
for _ in range(200):
    X, Y = rng.normal(size=6), rng.normal(size=6)
    errs.append(abs(B(X, Y) + (X[:3] @ Y[:3] + X[3:] @ Y[3:])))
check("H.B_equals_minus_uu_plus_vv", max(errs) < 1e-12, f"max err {max(errs):.2e}")
check("H.T_preserves_Pi", all(np.allclose(Pi(T(X)), Pi(X)) for X in rng.normal(size=(50, 6))))
check("H.T_reverses_h", all(np.allclose(h(T(X)), -h(X)) for X in rng.normal(size=(50, 6))))
check("H.joint_invariance_B(TX,TY)=B(X,Y)",
      max(abs(B(T(X), T(Y)) - B(X, Y)) for X, Y in rng.normal(size=(200, 2, 6))) < 1e-12)
sep = max(abs(B(T(X), Y) - B(X, Y)) for X, Y in rng.normal(size=(200, 2, 6)))
check("H.separate_invariance_FAILS", sep > 0.1, f"max |B(TX,Y)-B(X,Y)| = {sep:.3f}")
# kernel of Pi = {(u,-u)}; span{X-TX} equals it (dimension 3)
D = np.array([X - T(X) for X in rng.normal(size=(40, 6))])
rk = np.linalg.matrix_rank(D, tol=1e-9)
check("H.span{X-TX}_has_dim_3=dim_kerPi", rk == 3, f"rank {rk}")
check("H.span{X-TX}_inside_kerPi", max(np.abs(Pi(d)).max() for d in D) < 1e-12)
# witness: k=(u,-u), Y arbitrary => B(k,Y) = -1/2 h_k.h_Y != 0
k = np.concatenate([[1., 0, 0], [-1., 0, 0]]); Y = np.concatenate([[1., 0, 0], [-1., 0, 0]])
check("H.witness_Pi(k)=0_but_B(k,Y)=-2", np.allclose(Pi(k), 0) and abs(B(k, Y) + 2) < 1e-12, f"B={B(k,Y)}")
# obstruction ranks: rho_L = rank of X-> B(X,.) on ker Pi
K = np.array([np.concatenate([e, -e]) for e in np.eye(3)])
Gram = np.array([[B(k_, np.eye(6)[j]) for j in range(6)] for k_ in K])
check("H.obstruction_rank_rho_L=3", np.linalg.matrix_rank(Gram) == 3)

# ---------- Result I: factorization criterion (random finite-dim) ----------
def factors(Bm, P, tol=1e-9):
    """does Bm = P^T Bbar P for some Bbar ? test via kernel-in-radicals AND direct lstsq"""
    n = Bm.shape[0]
    Kk = np.linalg.svd(P)[2][P.shape[0]:].T if P.shape[0] < n else np.zeros((n, 0))
    krit = (Kk.size == 0) or (np.abs(Bm @ Kk).max() < tol and np.abs(Bm.T @ Kk).max() < tol)
    # direct: solve Bm = P^T Bbar P (P full row rank)
    Pp = np.linalg.pinv(P)
    Bbar = Pp.T @ Bm @ Pp
    direct = np.abs(P.T @ Bbar @ P - Bm).max() < tol
    return krit, direct
agree = True; n_fact = n_nonfact = 0
for trial in range(300):
    n, m = int(rng.integers(3, 8)), int(rng.integers(1, 3)) + 1
    m = min(m, n - 1)
    P = rng.normal(size=(m, n))
    if trial % 2 == 0: # construct a factoring B
        Bm = P.T @ rng.normal(size=(m, m)) @ P
    else: # generic non-factoring B
        Bm = rng.normal(size=(n, n))
    a, b = factors(Bm, P)
    agree &= (a == b); n_fact += a; n_nonfact += (not a)
check("ResultI.criterion_matches_direct_factorization(300 random)", agree,
      f"{n_fact} factoring / {n_nonfact} non-factoring cases")
# one-sided counterexample: kernel in Rad_L but not Rad_R => NOT factoring
n = 4; P = np.array([[1., 0, 0, 0], [0, 1, 0, 0]]); k = np.array([0, 0, 1., 0])
Bm = np.zeros((n, n)); Bm[0, 0] = 1; Bm[0, 2] = 1 # B(e0,e2)=1: e2 in ker; B(e2,.)=0 so in Rad_L; B(.,e2)!=0 so not Rad_R
check("ResultI.one_sided_radical_insufficient",
      np.abs(Bm.T @ k).max() < 1e-12 and np.abs(Bm @ k).max() > 0.5 and not factors(Bm, P)[1])

# ---------- Result III: involution / coinvariant, random ----------
ok = True
```

```
for trial in range(200):
    d1, d2 = int(rng.integers(1, 4)), int(rng.integers(1, 4))
    n = d1 + d2
    Tm = np.diag([1.]*d1 + [-1.]*d2) # involution, V+ (+) V-
    Q, _ = np.linalg.qr(rng.normal(size=(n, n))) # random basis change
    Ti = Q @ Tm @ Q.T
    Pm = (np.eye(n) + Ti) / 2 # projection onto V+ (observable)
    # ker Pi = V-, span{X-TX} = V-
    Bm = rng.normal(size=(n, n))
    # separately invariant projection: B -> Pm^T B Pm (the only separately invariant ones factor)
    Bs = Pm.T @ Bm @ Pm
    sepinv = np.abs(Ti.T @ Bs - Bs).max() < 1e-9 and np.abs(Bs @ Ti - Bs).max() < 1e-9
    # general joint-invariant: average
    Bj = (Bm + Ti.T @ Bm @ Ti) / 2
    jointinv = np.abs(Ti.T @ Bj @ Ti - Bj).max() < 1e-9
    # block diagonality: B(V+,V-)=0
    Vp = Q[:, :d1]; Vm = Q[:, d1:]
    blockdiag = np.abs(Vp.T @ Bj @ Vm).max() < 1e-9 and np.abs(Vm.T @ Bj @ Vp).max() < 1e-9
    # Result III: sep-inv => factors through Pi
    Pi_rows = Vp.T # Pi: V -> V+ coordinates
    f_sep = factors(Bs, Pi_rows)[0]
    # joint-inv with nonzero V- block does NOT factor
    nz = np.abs(Vm.T @ Bj @ Vm).max() > 1e-6
    f_joint = factors(Bj, Pi_rows)[0]
    ok &= sepinv and jointinv and blockdiag and f_sep and ((not f_joint) == nz)
check("ResultIII+Prop(block-diagonal): 200 random involutions", ok)

# ---------- Legendre family on the free space over S^2 ----------
def P(l, x):
    from numpy.polynomial import legendre as L
    c = np.zeros(l + 1); c[l] = 1; return L.legval(x, c)
# exact rational P3
P3 = lambda x: (5 * x**3 - 3 * x) / 2
half, sq3h_sq = Fr(1, 2), Fr(3, 4)
# exact: B3(X,r(ex)) = -P3(1/2) + (1/2) P3(1) + (sqrt3/2) P3(0); P3(0)=0 => sqrt3 term vanishes
B3_exact = -P3(half) + half * P3(Fr(1)) + 0
check("Legendre.P3(1/2)=-7/16", P3(half) == Fr(-7, 16), str(P3(half)))
check("Legendre.B3(X,r(ex))=15/16 exact", B3_exact == Fr(15, 16), str(B3_exact))
B1_exact = -half + half * 1 + 0
check("Legendre.B1(X,r(ex))=0 exact", B1_exact == 0)
check("Legendre.corr_60deg_B1=-1/2,B3=+7/16", -half == Fr(-1, 2) and -P3(half) == Fr(7, 16))
check("Legendre.P(++)_B1=1/8", (1 + Fr(-1, 2)) / 4 == Fr(1, 8))
check("Legendre.P(++)_B3=23/64=0.359375", (1 + Fr(7, 16)) / 4 == Fr(23, 64) and float(Fr(23, 64)) == 0.359375)
# numerical cross-check with unit vectors
n = np.array([.5, np.sqrt(3) / 2, 0]); ex = np.array([1., 0, 0]); ey = np.array([0, 1., 0])
Xv = [(1, n), (-.5, ex), (-np.sqrt(3) / 2, ey)]
for l in (1, 3):
    val = sum(-c * P(l, v @ ex) for c, v in Xv)
    print(f" numeric B_{l}(X,r(ex)) = {val:.12f}")
check("Legendre.numeric_matches", abs(sum(-c*P(3, v@ex) for c, v in Xv) - 15/16) < 1e-12 and abs(sum(-c*P(1, v@ex) for c, v in Xv)) < 1e
      -12)
# oddness of P_l(l odd): sign reversal property
check("Legendre.odd_l_sign_reversal", all(abs(P(l, -x) + P(l, x)) < 1e-12 for l in (1, 3, 5, 7) for x in np.linspace(-1, 1, 11)))
# obstruction rank on N generic points of S^2
def sphere_points(N, seed):
    r = np.random.default_rng(seed); Z = r.normal(size=(N, 3)); return Z / np.linalg.norm(Z, axis=1, keepdims=True)
N = 120; pts = sphere_points(N, 7)
Gm = pts @ pts.T
Pm_ = pts.T # 3 x N : Pi on coordinates r(a_i) -> a_i
Kb = np.linalg.svd(Pm_)[2][3:].T # N x (N-3) basis of ker Pi
ranks = {}
for l in range(1, 8):
    Bl = -P(l, np.clip(Gm, -1, 1))
    # sanity: PSD of -B
    ev = np.linalg.eigvalsh(-Bl); psd = ev.min() > -1e-9
    rho = np.linalg.matrix_rank(Kb.T @ Bl, tol=1e-8)
    ranks[l] = int(rho)
    print(f" l={l}: rank(B_l|ker Pi)={rho} (-B_l PSD: {psd})")
check("Legendre.B1_factors(rank 0)", ranks[1] == 0)
check("Legendre.B_l_l>=2_do_not_factor", all(ranks[l] > 0 for l in range(2, 8)))
check("Legendre.obstruction_rank=2l+1 (generic N=120)", all(ranks[l] == 2 * l + 1 for l in range(2, 8)), str(ranks))
out["obstruction_ranks_N120"] = ranks

# ---------- GCT v002 Theorem-3 symmetry conditions for c3(d) = -P3(cos d) ----------
ds = np.linspace(-np.pi, np.pi, 41)
c3 = lambda d: -P(3, np.cos(d))
cond = (abs(c3(0) + 1) < 1e-12 and all(abs(c3(-d) - c3(d)) < 1e-12 and abs(c3(d + np.pi) + c3(d)) < 1e-12 and abs(c3(d)) <= 1 + 1e-12 for
        d in ds))
check("GCTv002.Thm3_symmetry_conditions_hold_for_c3", cond)

# ---------- coinvariants: diagonal vs factorwise (Z2) ----------
# (V (x) V)_G has dim = dim(V+ (x) V+) + dim(V- (x) V-); V_G (x) V_G has dim = (dim V+)^2.
d1, d2 = 3, 3
check("Coinvariant.dim_gap=dim(V-)^2 for model H", (d1*d1 + d2*d2) - d1*d1 == d2*d2 == 9)


# ---------- Fig.1 concrete numbers (model H) ----------
Xa = np.array([1., 0, 0, 0, 0, 0]); Xb = T(Xa)
check("Fig1.same_observable_s=(1,0,0)", np.allclose(Pi(Xa), [1, 0, 0]) and np.allclose(Pi(Xb), [1, 0, 0]))
check("Fig1.B(X,X)=-1,B(TX,X)=0,B(TX,TX)=-1",
      abs(B(Xa, Xa) + 1) < 1e-12 and abs(B(Xb, Xa)) < 1e-12 and abs(B(Xb, Xb) + 1) < 1e-12)
# 2-D minimal example of Sec.1
```

```
Pi2 = lambda X: X[0]; T2 = lambda X: np.array([X[0], -X[1]]); B2 = lambda X, Y: X @ Y
X1, X2, Y1 = np.array([1., 1]), np.array([1., -1]), np.array([0., 1])
check("Minimal2D.values", Pi2(X1) == Pi2(X2) and B2(X1, Y1) == 1 and B2(X2, Y1) == -1 and B2(X1, X1) == B2(X2, X2) == 2)
# one-sided 2-D example B(X,Y)=x*y'
B3 = np.array([[0., 1], [0, 0]]) # B(X,Y)=X^T B3 Y = x*y'
Kk2 = np.array([[0.], [1.]])
check("OneSided2D.K_in_RadL_not_RadR", np.abs(Kk2.T @ B3).max() == 0 and np.abs(B3 @ Kk2).max() > 0)

# ---------- ell=5 exact ----------
P5 = lambda x: (63 * x**5 - 70 * x**3 + 15 * x) / 8
check("Legendre.l=5_exact", -P5(half) == Fr(-23, 256) and -P5(half) + half * P5(Fr(1)) == Fr(105, 256) and (1 + (-P5(half))) / 4 == Fr
      (233, 1024),
     f"C={-P5(half)}, B5(X,ex)={-P5(half)+half}, P(++)={(1-P5(half))/4}")

# ---------- relational deficit: formula and minimal completion (random) ----------
def deficit_formula(Bm, K):
    S = np.vstack([Bm @ K, Bm.T @ K]); return np.linalg.matrix_rank(S, tol=1e-9)
okd = True; cnt = 0
for trial in range(300):
    n = int(rng.integers(4, 9)); m = int(rng.integers(1, 3)) + 1
    Pm = rng.normal(size=(m, n)); Kk = np.linalg.svd(Pm)[2][m:].T
    Bm = rng.normal(size=(n, n))
    mode = trial % 3
    if mode == 1: # make part of K radical on both sides
        c = Kk[:, 0]; Bm = Bm - np.outer(Bm @ c, c) / (c @ c) - np.outer(c, c @ Bm) / (c @ c) + (c @ Bm @ c) * np.outer(c, c) / (c @ c)
            **2
    if mode == 2: # symmetric
        Bm = (Bm + Bm.T) / 2
    d = deficit_formula(Bm, Kk)
    rhoL = np.linalg.matrix_rank(Kk.T @ Bm, tol=1e-9); rhoR = np.linalg.matrix_rank(Bm @ Kk, tol=1e-9)
    okd &= (max(rhoL, rhoR) <= d <= rhoL + rhoR)
    if mode == 2: okd &= (d == rhoL == rhoR)
    # minimal completion Psi with delta rows
    S = np.vstack([Bm @ Kk, Bm.T @ Kk]); U, sv, Vt = np.linalg.svd(S, full_matrices=False)
    Q = Vt[:d] # d x dim K
    Psi = Q @ np.linalg.pinv(Kk) # d x n
    Pfull = np.vstack([Pm, Psi])
    okd &= factors(Bm, Pfull)[1]
    if d > 0 and Psi.shape[0] > 1: # delta-1 rows must fail
        okd &= not factors(Bm, np.vstack([Pm, Psi[:-1]]))[1]
    cnt += 1
check("Deficit.bounds+formula+minimal_completion (300 random)", okd, f"{cnt} cases")

# ---------- permutation example: joint vs separate invariant forms on R^n ----------
n = 5
def invariant_dim(joint):
    rows = []
    for i in range(n - 1):
        Pm = np.eye(n); Pm[[i, i + 1]] = Pm[[i + 1, i]]
        # constraint on vec(B): joint: P^T B P = B ; separate: P^T B = B and B P = B
        I = np.eye(n)
        if joint: rows.append(np.kron(Pm.T, Pm.T) - np.eye(n * n)) # vec(P^T B P)=(P^T (x) P^T)vec(B) for row-major
        else:
            rows.append(np.kron(Pm.T, I) - np.eye(n * n)); rows.append(np.kron(I, Pm.T) - np.eye(n * n))
    A = np.vstack(rows); return n * n - np.linalg.matrix_rank(A, tol=1e-9)
check("Perm.joint_invariant_forms_dim=2,separate=1", invariant_dim(True) == 2 and invariant_dim(False) == 1,
      f"joint {invariant_dim(True)}, separate {invariant_dim(False)}")
Pm = np.ones((1, n)); Kk = np.linalg.svd(Pm)[2][1:].T
Bperm = np.eye(n) + 0.3 * np.ones((n, n))
check("Perm.deficit=n-1=4", deficit_formula(Bperm, Kk) == n - 1)
# span of {X - T_i X} = sum-zero hyperplane (H2)
D = []
for i in range(n - 1):
    Pt = np.eye(n); Pt[[i, i + 1]] = Pt[[i + 1, i]]
    D += [(np.eye(n) - Pt)[:, j] for j in range(n)]
check("Perm.H2_span{X-T_iX}=ker(sum) (dim 4)", np.linalg.matrix_rank(np.array(D), tol=1e-9) == n - 1)

# ---------- Closure illustration: ideal Helmholtz projector per Fourier mode ----------
kv = np.array([1., 2.]); Pk = np.eye(2) - np.outer(kv, kv) / (kv @ kv); Rk = 2 * Pk - np.eye(2)
Bk = np.eye(2)
okc = np.allclose(Pk @ Pk, Pk) and np.allclose(Pk @ kv, 0) and np.allclose(Rk @ Rk, np.eye(2)) and np.allclose(Pk @ Rk, Pk)
okc &= np.allclose(Rk.T @ Bk @ Rk, Bk) and not np.allclose(Rk.T @ Bk, Bk)
check("Closure.ideal_Helmholtz_per_mode: P idem, R involution, joint-inv, not separate", okc)
check("Closure.deficit=1_per_mode", deficit_formula(Bk, kv.reshape(2, 1)) == 1)


# ---------- four-route table (Example 5.x): exact arithmetic in Q(sqrt3) ----------
class QS:
    """a + b*sqrt(3), a,b rational (Fractions)."""
    def __init__(s, a=0, b=0): s.a, s.b = Fr(a), Fr(b)
    def __add__(s, o): o = o if isinstance(o, QS) else QS(o); return QS(s.a + o.a, s.b + o.b)
    __radd__ = __add__
    def __neg__(s): return QS(-s.a, -s.b)
    def __sub__(s, o): return s + (-(o if isinstance(o, QS) else QS(o)))
    def __mul__(s, o): o = o if isinstance(o, QS) else QS(o); return QS(s.a * o.a + 3 * s.b * o.b, s.a * o.b + s.b * o.a)
    __rmul__ = __mul__
    def __eq__(s, o): o = o if isinstance(o, QS) else QS(o); return s.a == o.a and s.b == o.b
    def __repr__(s): return f"({s.a}+{s.b}*sqrt3)"
Z, O1, H2, S3 = QS(0), QS(1), QS(Fr(1, 2)), QS(0, Fr(1, 2)) # 0, 1, 1/2, sqrt3/2
dirs4 = [[O1, Z, Z], [Z, O1, Z], [Z, Z, O1], [H2, S3, Z]]
dot = lambda u, v: u[0] * v[0] + u[1] * v[1] + u[2] * v[2]
P3q = lambda t: (5 * t * t * t - 3 * t) * Fr(1, 2)
```

```
G1 = [[-dot(dirs4[i], dirs4[j]) for j in range(4)] for i in range(4)]
G3 = [[-P3q(dot(dirs4[i], dirs4[j])) for j in range(4)] for i in range(4)]
kq = [QS(-Fr(1, 2)), QS(0, -Fr(1, 2)), Z, O1] # (-1/2, -sqrt3/2, 0, 1)
matvec = lambda G, v: [sum((G[i][j] * v[j] for j in range(4)), QS(0)) for i in range(4)]
Pik = [sum((dirs4[j][c] * kq[j] for j in range(4)), QS(0)) for c in range(3)]
check("FourRoute.k_in_kerPi", all(p == 0 for p in Pik))
check("FourRoute.G1*k=0", all(v == 0 for v in matvec(G1, kq)))
g3k = matvec(G3, kq)
exp3 = [QS(Fr(15, 16)), QS(0, Fr(5, 16)), QS(0), QS(-Fr(15, 16))]
check("FourRoute.G3*k=(15/16,5sqrt3/16,0,-15/16)", all(g3k[i] == exp3[i] for i in range(4)), str(g3k))
check("FourRoute.G3_entries", G3[0][3] == QS(Fr(7, 16)) and G3[1][3] == QS(0, -Fr(3, 16)) and G3[0][1] == 0 and G3[2][3] == 0
      and all(G3[i][i] == QS(-1) for i in range(4)))

G3f = np.array([[float(G3[i][j].a) + float(G3[i][j].b) * np.sqrt(3) for j in range(4)] for i in range(4)])
G1f = np.array([[float(G1[i][j].a) + float(G1[i][j].b) * np.sqrt(3) for j in range(4)] for i in range(4)])
kf = np.array([float(v.a) + float(v.b) * np.sqrt(3) for v in kq]).reshape(4, 1)
check("FourRoute.deficit_B3=1,B1=0", deficit_formula(G3f, kf) == 1 and deficit_formula(G1f, kf) == 0)

# ---------- self-interaction descends iff symmetric B descends ----------
okq = True
for trial in range(100):
    n = int(rng.integers(4, 8)); m = int(rng.integers(1, 3)) + 1
    Pm = rng.normal(size=(m, n)); Kk = np.linalg.svd(Pm)[2][m:].T
    if trial % 2 == 0: Bm = Pm.T @ rng.normal(size=(m, m)) @ Pm; Bm = (Bm + Bm.T) / 2 # factoring, symmetric
    else: Bm = rng.normal(size=(n, n)); Bm = (Bm + Bm.T) / 2
    Q = lambda z: z @ Bm @ z
    Xs = rng.normal(size=(8, n)); ks = (Kk @ rng.normal(size=(Kk.shape[1], 8))).T
    qdesc = all(abs(Q(xv + kv) - Q(xv)) < 1e-9 for xv, kv in zip(Xs, ks))
    okq &= (qdesc == factors(Bm, Pm)[1])
check("SelfInteraction.Q_descends_iff_B_descends (100 random symmetric)", okq)

# ---------- representatives: induced pairing on O depends on the section unless K subset Rad ----------
okr = True
for trial in range(100):
    H = rng.normal(size=(3, 3))
    # section sigma_H(s) = (s, H s) in (s,h) coordinates of model H; B in (s,h)-coords = -1/2 (s.s' + h.h')
    BsH = -0.5 * (np.eye(3) + H.T @ H)
    X = np.concatenate([[1., 2, 3], H @ np.array([1., 2, 3])]); Y = np.concatenate([[0., 1, -1], H @ np.array([0., 1, -1])])
    okr &= abs(-0.5 * (X[:3] @ Y[:3] + X[3:] @ Y[3:]) - (np.array([1., 2, 3]) @ BsH @ np.array([0., 1, -1]))) < 1e-10
check("Sections.model_H_induced_pairing=-1/2(I+H^T H)", okr)

# ---------- Section 8: generative response profiles ----------
def nullspace(M, tol=1e-9):
    if M.shape[0] == 0: return np.eye(M.shape[1])
    _, sv, vt = np.linalg.svd(M); r = int((sv > tol).sum()); return vt[r:].T
def largest_invariant_in(W, F):
    cur = W
    for _ in range(F.shape[0] + 2):
        if cur.shape[1] == 0: return cur
        Pperp = np.eye(F.shape[0]) - cur @ cur.T # cur has orthonormal columns
        Nn = nullspace(Pperp @ F @ cur)
        new = cur @ Nn if Nn.shape[1] else np.zeros((F.shape[0], 0))
        if new.shape[1] == cur.shape[1]: return new
        cur = new
    return cur
def same_subspace(A, Bm):
    if A.shape[1] != Bm.shape[1]: return False
    if A.shape[1] == 0: return True
    PA = A @ np.linalg.pinv(A); PB = Bm @ np.linalg.pinv(Bm); return np.allclose(PA, PB, atol=1e-7)
okS = okC = True; nS = 0
for trial in range(300):
    n = int(rng.integers(4, 8)); m = int(rng.integers(1, 3))
    Pm = rng.normal(size=(m, n)); Kb = nullspace(Pm)
    mode = trial % 3
    Bm = rng.normal(size=(n, n))
    if mode == 1: Bm = Pm.T @ rng.normal(size=(m, n)) # K inside Rad_L (left descends)
    F = rng.normal(size=(n, n)) * 0.7 if trial % 2 == 0 else np.eye(n) + Kb @ rng.normal(size=(Kb.shape[1], n)) # general / observable-
        preserving
    stack = np.vstack([np.vstack([Pm @ np.linalg.matrix_power(F, k), Bm.T @ np.linalg.matrix_power(F, k)]) for k in range(n)])
    SG = nullspace(stack) # profile congruence subspace (direct)
    W0 = nullspace(np.vstack([Pm, Bm.T])) # K cap Rad_L
    okS &= same_subspace(SG, largest_invariant_in(W0, F))
    # completeness: K subset S_G <=> K invariant under F and K subset Rad_L
    KinSG = np.allclose(SG @ np.linalg.pinv(SG) @ Kb, Kb, atol=1e-7) if SG.shape[1] else (Kb.shape[1] == 0)
    FKinK = np.allclose(Kb @ np.linalg.pinv(Kb) @ F @ Kb, F @ Kb, atol=1e-7)
    KinRadL = np.abs(Bm.T @ Kb).max() < 1e-9
    okC &= (KinSG == (FKinK and KinRadL))
    if trial % 2 == 1: okC &= (FKinK and (KinSG == KinRadL)) # observable-preserving => K invariant
    nS += 1
check("Profile.congruence_subspace=largest_F-invariant_subspace_of_K_cap_RadL (300 random)", okS, f"{nS} cases")
check("Profile.completeness: ~Pi => ~G iff K F-invariant and K in Rad_L (300 random)", okC)

# hidden-sector model vs first-order model under T
PiH = np.hstack([np.eye(3), np.eye(3)]); TH = np.block([[np.zeros((3, 3)), np.eye(3)], [np.eye(3), np.zeros((3, 3))]])
B_H = -np.eye(6); B_1 = -PiH.T @ PiH
def SG_dim(Pm, Bm, F):
    n = F.shape[0]; st = np.vstack([np.vstack([Pm @ np.linalg.matrix_power(F, k), Bm.T @ np.linalg.matrix_power(F, k)]) for k in range(n)
        ])
    return nullspace(st).shape[1]
check("Profile.hidden-sector: S_G=0 (route fully recoverable from profile); first-order: S_G=K (dim 3)",
     SG_dim(PiH, B_H, TH) == 0 and SG_dim(PiH, B_1, TH) == 3)

```

```
# identifiability: probes needed = rho_L ; compatible class dimension after m probes
okI = True
for trial in range(200):
    n = int(rng.integers(4, 9)); m = int(rng.integers(1, 3)); Pm = rng.normal(size=(m, n)); Kb = nullspace(Pm); k = Kb.shape[1]
    Bm = rng.normal(size=(n, n))
    if trial % 2: Bm = Bm - (Kb[:, :1] @ (Kb[:, :1].T @ Bm)) # make one K-direction left-radical
    rhoL = np.linalg.matrix_rank(Kb.T @ Bm, tol=1e-9)
    def dimKJ(mp):
        if mp == 0: return k
        Eta = rng.normal(size=(n, mp)); R = (Kb.T @ Bm @ Eta).T # rows: probe functionals restricted to K
        return k - np.linalg.matrix_rank(R, tol=1e-9)
    okI &= dimKJ(rhoL) == k - rhoL and (rhoL == 0 or dimKJ(rhoL - 1) > k - rhoL)
    # full protocol residual = dim(K cap Rad_L)
    okI &= (k - rhoL) == nullspace(np.vstack([Pm, Bm.T])).shape[1]
check("Identifiability: rho_L generic probes resolve the fibre up to K cap Rad_L; rho_L-1 do not (200 random)", okI)

# admissible fibres: K_A and observable adequacy on A
okA = True
for trial in range(200):
    n = int(rng.integers(5, 9)); m = 2; Pm = rng.normal(size=(m, n)); Kb = nullspace(Pm)
    u = Kb[:, 0]
    base = rng.normal(size=(4, n)); A = []
    for x0 in base:
        A += [x0, x0 + rng.normal() * u, x0 + rng.normal() * u] # admissible routes: fibres only along u
    A = np.array(A)
    D = [A[i] - A[j] for i in range(len(A)) for j in range(len(A)) if i != j and np.abs(Pm @ (A[i] - A[j])).max() < 1e-9]
    KA = np.array(D)
    Bm = rng.normal(size=(n, n))
    if trial % 2 == 0: Bm = Bm - np.outer(u, u @ Bm) # u in Rad_L, other K directions not
    adequate = all(abs((A[i] - A[j]) @ Bm @ eta) < 1e-9 for i in range(len(A)) for j in range(len(A))
                   if np.abs(Pm @ (A[i] - A[j])).max() < 1e-9 for eta in np.eye(n))
    KA_in_RadL = np.abs(KA @ Bm).max() < 1e-9
    okA &= (adequate == KA_in_RadL) and (np.linalg.matrix_rank(KA, tol=1e-9) <= 1) and (trial % 2 == 1 or KA_in_RadL)
    okA &= (np.abs(Kb.T @ Bm).max() > 1e-9) # but K itself is NOT inside Rad_L
check("AdmissibleFibres: adequacy on A iff K_A in Rad_L, with K_A a proper subspace of K (200 random)", okA)


# ---- Section 8 (continued): worked examples and gauge invariance ----
# (a) the (u,v) witness: X=(e_x,0), X'=(0,e_x), Y=(e_x,0) with B=-(u.a+v.b)
ex = np.array([1., 0, 0]); z3 = np.zeros(3)
Xa, Xb, Yc = np.concatenate([ex, z3]), np.concatenate([z3, ex]), np.concatenate([ex, z3])
Bm_uv = -np.eye(6)
check("Witness.same_Pi_different_B: Pi(X)=Pi(X')=e_x, B(X,Y)=-1, B(X',Y)=0",
      np.allclose(PiH @ Xa, PiH @ Xb) and np.isclose(Xa @ Bm_uv @ Yc, -1) and np.isclose(Xb @ Bm_uv @ Yc, 0))
# (b) two-dimensional example: dynamics reveals a distinction invisible to Pi and B
P2 = np.array([[1., 0.]]); B2 = np.array([[1., 0.], [0., 0.]]); F1 = np.eye(2); F2 = np.array([[1., 1.], [0., 1.]])
def prof(Pm, Bm, F, X, N=4):
    return [np.concatenate([Pm @ np.linalg.matrix_power(F, k) @ X, (np.linalg.matrix_power(F, k) @ X) @ Bm]) for k in range(N)]
Z = np.array([0., 1.]) # Z in K = ker Pi, and in Rad_L(B)
d_zero = np.abs(B2.T @ Z).max() < 1e-12 and np.abs(P2 @ Z).max() < 1e-12
n1 = max(np.abs(p).max() for p in prof(P2, B2, F1, Z)); n2 = [np.abs(p).max() for p in prof(P2, B2, F2, Z)]
check("Example.dynamics_separates: Z in K cap Rad_L; M1 (F=id) cannot separate Z from 0, M2 (F=shear) does at n>=1",
      d_zero and n1 < 1e-12 and n2[0] < 1e-12 and all(v > 1e-9 for v in n2[1:]) and SG_dim(P2, B2, F1) == 1 and SG_dim(P2, B2, F2) == 0)
# (c) gauge invariance under g with Pi g = Pi
okG = True
for trial in range(200):
    n = int(rng.integers(4, 8)); m = int(rng.integers(1, 3)); Pm = rng.normal(size=(m, n)); Kb = nullspace(Pm)
    Bm = rng.normal(size=(n, n)); Bm = Bm - Kb[:, :1] @ (Kb[:, :1].T @ Bm) if trial % 2 else Bm
    F = rng.normal(size=(n, n)) * 0.7
    g = np.eye(n) + Kb @ rng.normal(size=(Kb.shape[1], n)) * 0.3
    if abs(np.linalg.det(g)) < 1e-3: continue
    gi = np.linalg.inv(g); B_ = gi.T @ Bm @ gi; F_ = g @ F @ gi
    okG &= np.allclose(Pm @ g, Pm)
    X = rng.normal(size=n); eta = rng.normal(size=n)
    for k in range(4):
        Fk = np.linalg.matrix_power(F, k); F_k = np.linalg.matrix_power(F_, k)
        okG &= np.allclose(Pm @ F_k @ (g @ X), Pm @ Fk @ X) and np.isclose((F_k @ (g @ X)) @ B_ @ (g @ eta), (Fk @ X) @ Bm @ eta)
    def dl(Pm, Bm):
        Kc = nullspace(Pm); return Kc.shape[1] - nullspace(np.vstack([Pm, Bm.T])).shape[1]
    okG &= SG_dim(Pm, Bm, F) == SG_dim(Pm, B_, F_) and dl(Pm, Bm) == dl(Pm, B_)
    okG &= np.linalg.matrix_rank(np.vstack([Kb.T @ Bm, Kb.T @ Bm.T]), tol=1e-9) == np.linalg.matrix_rank(np.vstack([Kb.T @ B_, Kb.T @ B_.
        T]), tol=1e-9)
check("Gauge: g with Pi g = Pi relabels routes but preserves every protocol output; dim S_G, rho_L, delta invariant (200 random)", okG)
# (d) trivial realisation: V=0, Pi=id gives K=0, delta=0, S_G=0
check("Weak_claim.trivial_realisation: Pi=id has K=0, delta=0, S_G=0 for any B,F", all(
    SG_dim(np.eye(k), rng.normal(size=(k, k)), rng.normal(size=(k, k))) == 0 for k in (2, 3, 5)))


# ---- v2 review regressions ----
# (A) Prop 4.9(c) needs O != 0: V=R, O=0, B=xy has a unique section yet does not descend
K_A = np.array([[1.]]); B_A = np.array([[1.]])
check("Regression.Prop4.9c: O=0 counterexample (unique section, B not descending, K not in Rad)",
      np.abs(B_A @ K_A).max() > 0 and K_A.shape[1] == 1)
# (B) fixed-label vs relabelled outputs under g=(x,2y)
Bq = np.array([[0., 0.], [0., 1.]]); g2 = np.diag([1., 2.]); gi2 = np.linalg.inv(g2); Bg = gi2.T @ Bq @ gi2; ey = np.array([0., 1.])
check("Regression.Prop8.8: fixed-label output 1 vs 1/4; relabelled (g e_y, g e_y) output 1",
      np.isclose(ey @ Bq @ ey, 1) and np.isclose(ey @ Bg @ ey, 0.25) and np.isclose((g2 @ ey) @ Bg @ (g2 @ ey), 1))
# (C) gauge fixing named routes/probes leaves all fixed-label outputs unchanged
okF = True
for trial in range(200):
    n = int(rng.integers(4, 8)); m = int(rng.integers(1, 3)); Pm = rng.normal(size=(m, n)); Kb = nullspace(Pm)
    Bm = rng.normal(size=(n, n)); F = rng.normal(size=(n, n)) * 0.7
```

```
    X = rng.normal(size=n); eta = rng.normal(size=n)
    Q, _ = np.linalg.qr(np.stack([X, eta], 1)); proj = np.eye(n) - Q @ Q.T
    g = np.eye(n) + Kb @ rng.normal(size=(Kb.shape[1], n)) @ proj * 0.3
    if abs(np.linalg.det(g)) < 1e-3: continue
    gi = np.linalg.inv(g); B_ = gi.T @ Bm @ gi; F_ = g @ F @ gi
    okF &= np.allclose(g @ X, X) and np.allclose(g @ eta, eta)
    for k in range(4):
        okF &= np.allclose(Pm @ np.linalg.matrix_power(F_, k) @ X, Pm @ np.linalg.matrix_power(F, k) @ X)
        okF &= np.isclose((np.linalg.matrix_power(F_, k) @ X) @ B_ @ eta, (np.linalg.matrix_power(F, k) @ X) @ Bm @ eta)
check("Regression.Prop8.8c: g fixing named routes and probes leaves fixed-label outputs unchanged (200 random)", okF)


# ---- Section 10.1: canonical nest operation M (x) N = MN + M/N + 2M ----
nest = lambda M, N: Fr(M * N) + Fr(M, N) + 2 * M
check("Canonical.nest_identity: M(x)N = M(N+1)^2/N (59x59 exact)", all(nest(M, N) == Fr(M * (N + 1) ** 2, N) for M in range(1, 60) for N
      in range(1, 60)))
fib72 = [(M, N) for N in range(1, 71) for M in range(1, 73) if nest(M, N) == 72]
check("Canonical.fibre_of_72: exactly (18,1),(16,2),(10,5), all anchors M=kN with k(N+1)^2=72",
     sorted(fib72) == [(10, 5), (16, 2), (18, 1)] and all(M % N == 0 and (M // N) * (N + 1) ** 2 == 72 for M, N in fib72)
     and all(Fr(72 * N, (N + 1) ** 2) < 1 for N in range(70, 400)))
fibs = {}
for N in range(1, 41):
   for M in range(1, 201): fibs.setdefault(nest(M, N), []).append((M, N))
check("Canonical.non_injective: 235 values in (M<=200,N<=40) have >=2 routes", sum(1 for v in fibs.values() if len(v) > 1) == 235)

json.dump(out, open("verify_ri_report.json", "w"), indent=1)
print("\nALL PASS" if ok_all else "\nSOME FAILED"); sys.exit(0 if ok_all else 1)
```

C Source Provenance

C.1 Files consulted

The following files of the UNNS_ENTANGLE_I project were read and are cited; none was modified. SHA-256 digests are of the files as read on 21 October 2026.

Key	File	SHA-256
G1	docs/GCT_AXIOMS_v001.md	5df0c5478a8fe294053fb6613f315a44c280bc1a16ee37978828f9349ded33ac
G2	docs/GCT_JOINT_THEOREM_v002.md	2f54d4aba6dafebb5e3a1d3fd4effd819e72e0c3307cd648ea40e117e77e28bf
G3	docs/GCT_GENERATIVE_CONSTRUCTION_v002.md	833fdf0a2aa73012bda38616b7571ae415d79f468d7537125e74a1f214c2acb8
G4	docs/GCT_VERDICT_v002.md	9d291c4c10244fbcae086c9980a57699f6ee8ca3367e4d4e79554b4344cb90d7
G4	docs/GCT_ADVERSARIAL_AUDIT_v002.md	c3815356a8783805465e077710ea27661dac76265e0d558d5de9559a6bb13dd5
G5	docs/XVI_SOURCE_AUDIT_v001.md	1cc02785a885784af73f15cdb371a9d3897fe99c93157eb7e94f42981fc82853
G6	docs/GCT_VERDICT_v001.md	2879e300093937d2a90939b1bde1be477d71f3bacb9dc769121a2755bc75072f
G7	docs/sources/XVI_ENGINE_v084.html	da4f217eaba86eba02517e6f6c2d70d8e5da43b189158b3ad22b93399d47d44d
G8	config/GCT_LOCK_v002.json	014a58c3b9fef6ca3391bac965c73537d661af82eb2edbc00401c63c884c7d66
G9	relational_invariants.pdf	3dd3b21a12e3d59101b89ffb3aa505bb9ebbbba4064ab08e4eb9934056e6e9b4
G10	36ead142-attachment.txt	b1f552af73137b82c5bdb946fe415321960d15c067c8fcbd84e4fc55e1ae5cc2

G10 is a discussion text supplied with the task for version 2, not a file of the project.

G11 (when_a_number_becomes_a_route.html) is a public-facing article held in the UNNS Substrate project store (creation stamp 3 July 2026), read on 21 October 2026; no digest was computed for it. Also read but not cited: docs/GCT_THEOREM_v001.md, docs/GCT_BELL_AUDIT_

`v001.md`, `docs/OPERATOR_MAP.md`, `docs/CONTEXT.md`, `GCT_README_v002.md`. The Chamber XVI HTML supplied with this task (`UNNS-Chamber_XVI_v0.8.4_STABLE.html`) has the same SHA-256 digest as G7, so the two are byte-identical.

C.2 What each source contributed

Source	Used for
G1, G6	the v001 frozen assumptions and the no-go verdict under Bell-local factorisation (context only)
G2, G8	binary joint-event representation; symmetry conditions on the kernel; the underdetermination family (context; Remark 5.7; arithmetic in Table 3)
G3	vocabulary: alphabet, generation, constraints, ordering; the weak sense of admissibility; the statement that the grammar is a candidate, not canonical
G4	the v002 verdict and the stopping rule
G5, G7	description of Operator XVI as a Helmholtz decomposition; audit notes on the discrete implementation (§10.4)
G9	the route-composition analysis in which the Legendre counterexample, the observable-null refinement condition and the (s, h) decomposition first appear
G11	the canonical objects of §10.1: the nest operation, the generated occurrence (rule, position, route, boundary, status), the five admissibility criteria, the substrate tuple, and the remark that two occurrences of one numeral can differ in status
G10	the proposals formalised in §8: the mechanism $M = (\Sigma, R, C, O)$, the response profile, observational equivalence of mechanisms, the four criteria, the three identity levels, the weak and strong claims and the statement of Principle 8.14

C.3 Introduced versus adopted, and limits of the record

All formal definitions and results of Sections 2–7 are statements about the linear route systems of Definition 2.1. The Legendre counterexample and the hidden-sector decomposition are *reconstructed and verified independently* here from their defining formulae as they appear in G9; the theorem statements, the deficit, the Refinement-Compatibility Theorem with its hypotheses, the permutation example and all proofs were written for this manuscript. The test package that G9 reports for its own two-state recurrent route realisation ("GCT v003") is not part of the files provided and is neither reproduced nor relied upon (§9). The proposals of §8 appear in G10 in prose; every proposition, proof, example and test of that section was written for this manuscript, and G10's definitions were restricted to the linear setting, where they are the stated objects (G10 does not assert them as established). No reference to the UNNS program bibliography is made, because none of its entries is needed for the mathematics.

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[12] B. S. Tsirelson, Quantum generalizations of Bell's inequality, *Lett. Math. Phys.* **4** (1980), 93–100.

[13] UNNS Entanglement project (`UNNS_ENTANGLE_I`), *GCT Axioms v001*, `docs/GCT_AXIOMS_v001.md`.

[14] *GCT Joint Correlation Representation Theorem v002*, `docs/GCT_JOINT_THEOREM_v002.md`.

[15] *GCT v002 — Joint Generative Structure*, `docs/GCT_GENERATIVE_CONSTRUCTION_v002.md`.

[16] *GCT v002 Scientific Verdict* and *Adversarial Audit and Frozen Stopping Rule*, `docs/GCT_VERDICT_v002.md`, `docs/GCT_ADVERSARIAL_AUDIT_v002.md`.

[17] *Canonical XVI source reconciliation*, `docs/XVI_SOURCE_AUDIT_v001.md`.

[18] *GCT Verdict v001*, `docs/GCT_VERDICT_v001.md`.

[19] *UNNS Laboratory — Chamber XVI (Closure)*, Operator XVI, engine v0.8.4, `docs/sources/XVI_ENGINE_v084.html`.

[20] *GCT v002 lock*, `config/GCT_LOCK_v002.json`.

[21] *GCT-III — Generative Bilinearity: The Route Test* and the follow-up analysis of observable identity, transcript `relational_invariants.pdf` (15 pp.), provided with this task.

[22] *When a Number Becomes a Route: Entering the UNNS Substrate*, unns.tech article, UNNS Substrate Research Program, 2026 (project document `when_a_number_becomes_a_route.html`); it cites the manuscript *Introduction to the UNNS Substrate*, which was not consulted.

[23] Discussion text on generative characterisation and response profiles, `36ead142-attachment.txt`, supplied with this task.